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3. AIR QUALITY

3.1 Introduction

3.1.1.1 This section presents the assessment on potential air quality impacts on the air sensitive receivers (ASRs) arising from construction and operation of the San Tin Section (hereinafter as the "ST Section") of the Northern Metropolis Highway (NMH). Assessment has been conducted in accordance with the criteria and guidelines as stipulated in Annex 4 and Annex 12 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) as well as the requirements given in Clause 3.5.3 and Appendix B of the EIA Study Brief (ESB-373/2025).

3.2 Environmental Legislation, Standards, and Guidelines

3.2.1.1 The Air Pollution Control Ordinance (APCO) provides the statutory authority for controlling air pollutants from a variety of sources. The Hong Kong Air Quality Objectives (AQOs), which stipulate the maximum allowable concentrations over specific periods for typical pollutants, should be met. The prevailing AQOs are listed in **Table 3.1**.

Table 3.1 Hong Kong Air Quality Objectives

Pollutants	Averaging Time	Concentration Limit (μg/m³) [1]	No. of Exceedances to be Allowed per Calendar Year
Sulphur Dioxide (SO_2)	10-min	500	3
	24-hour	40	3
Respirable Suspended Particulates (PM_{10} / RSP) [2]	24-hour	75	9
	Annual	30	Not applicable
Fine Suspended Particulates ($PM_{2.5}$ / FSP) [3]	24-hour	37.5	18
	Annual	15	Not applicable
Nitrogen Dioxide (NO_2)	1-hour	200	18
	24-hour	120	9
	Annual	40	Not applicable
Ozone (O_3)	8-hour	160	9
	Peak Season	100	Not applicable
Carbon Monoxide (CO)	1-hour	30000	0
	8-hour	10000	0
	24-hour	4000	0
Lead (Pb)	Annual	0.5	Not applicable

Notes:

[1] All measurements of the concentration of gaseous air pollutants, i.e. sulphur dioxide, nitrogen dioxide, ozone and carbon monoxide, are to be adjusted to a reference temperature of 293 K and a reference pressure of 101.325 kPa.

[2] Respirable suspended particulates mean suspended particles in air with a nominal aerodynamic diameter of 10 μm or less.

[3] Fine suspended particulates mean suspended particles in air with a nominal aerodynamic diameter of 2.5 μm or less.

Air Pollution Control (Construction Dust) Regulation

3.2.1.2 Notifiable and regulatory works are under the control of Air Pollution Control (Construction Dust) Regulation. Notifiable works include site formation, reclamation, demolition, foundation and superstructure construction for buildings and road construction. Regulatory works include building renovation, road opening and resurfacing, slope stabilisation, and other activities including stockpiling, dusty material handling, excavation, concrete production, etc. This Project is expected to include notifiable works (foundation and superstructure construction and demolition) and regulatory works (dusty material handling and excavation). Contractors and site agents are required to inform Environmental Protection Department (EPD) and adopt air quality control measures to minimise air emission, while carrying out construction works, to the acceptable level.

Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation

3.2.1.3 The Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation comes into effect on 1 June 2015. Under the Regulation, Non-road mobile machinery (NRMM), except those exempted, are required to comply with the prescribed emission standards. From 1 September 2015, all regulated machines sold or leased for use in Hong Kong must be approved or exempted with a proper label in a prescribed format issued by EPD. Starting from 1 December 2015, only approved or exempted NRMM with a proper label are allowed to be used in specified activities and locations including construction sites. The Contractor is required to ensure the adopted machines or non-road vehicle under the Project could meet the prescribed emission standards and requirement.

Air Pollution Control (Fuel Restriction) (Amendment) Regulation

3.2.1.4 The Air Pollution Control (Fuel Restriction) (Amendment) Regulation 2024 was to impose legal control on the type of fuels allowed for use and their sulphur contents in commercial and industrial processes to reduce sulphur dioxide (SO_2) emissions. This regulation was commenced on 1 April 2025 and the sulphur content of liquid fuel has been tightened to 0.001% by weight.

Development Bureau Technical Circular (Works)

3.2.1.5 Development Bureau Technical Circular (Works) (DEVB TC(W)) No. 13/2020 Timely Application of Temporary Electricity and Water Supply for Public Works Contract and Wider Use of Electric Vehicles in Public Works Contracts is one of the environmental guidelines on timely application of temporary electricity and wider use of electric vehicles (EVs) in public works contract. The Project team should also specify the use of EV(s) and installation of a designated medium-speed charger for each EV in each public contract.

3.2.1.6 The Development Bureau Technical Circular (Works) No. 1/2015 Emissions Control of Non-Road Mobile Machinery in Capital Works Contracts of Public Works also requires that no exempted generators, air compressors, excavators and crawler cranes shall be allowed in the new capital works contracts of public works (including design and build contracts) with an estimated contract value exceeding $200 million, unless is at the discretion of the Architect/Engineer considering no feasible alternative.

Practice Note on Control of Air Pollution in Vehicle Tunnels

3.2.1.7 The Practice Note on Control of Air Pollution in Vehicle Tunnels prepared by EPD provides guidelines on control of air pollution in vehicle tunnels. Guideline values on tunnel air quality are presented in **Table 3.2** .

Table 3.2 Tunnel Air Quality Guidelines

Pollutant	Averaging Time	Maximum Concentration	
		µg/m³ [1]	ppm
Carbon Monoxide (CO)	5 minutes	115,000	100
Nitrogen Dioxide (NO_2)	5 minutes	1,800	1
Sulphur Dioxide	5 minutes	1,000	0.4

Note:
[1] Measured at 298K and 101.325kPa.

3.3 Description of the Environment

3.3.1.1 The ST Section of the NMH links the interchange at the San Tin Highway and traverses the developed area in Mo Fan Heung, which are presently brownfield sites. It passes through the conservation areas near Fung Kat Heung and Ngau Tam Mei, goes underground at Ngau Tam Shan. The alignment travels along the villages near Saddle Pass, extends through the conservation areas near San Tin, continues underground at Hadden Hill, spans across the Fanling Highway (Pak Shek Au Section), and finally connects to the Kwu Tung North New Development Area. The surrounding land uses are mainly residential areas, villages, government, institution or community (G/IC) uses, and educational institution.

3.3.1.2 The nearest EPD air quality monitoring station (AQMS) is Yuen Long AQMS. The recent five years (2020 – 2024) air pollutant concentrations relevant to the ST Section in Yuen Long AQMS are summarized in **Table 3.3**. Another nearby EPD AQMS is North AQMS, located close to the ST Section near Pak Shek Au. **Table 3.4** presents the air pollutant concentrations in North AQMS from the recent five years (2020 – 2024).

3.3.1.3 Exceedances were recorded at the Yuen Long AQMS for the 10th highest 24-hour RSP levels, as well as the 19th highest 24-hour and annual FSP levels. At the North AQMS, the annual FSP level marginally exceeded the AQO. Despite these exceedances, a general decreasing trend in air pollutant concentrations was observed.

Table 3.3 Average Concentrations of Pollutants in the Recent Five Years (Year 2020 – 2024) at Yuen Long Air Quality Monitoring Station

Pollutants	Averaging Time		AQOs (µg/m³)	2020	2021	2022	2023	2024
RSP	24-hr	10th Highest	75	**77**	73	56	59	61
	Annual		30	30	30	25	26	25
FSP	24-hr	19th Highest	37.5	33	36	**38**	34	**38**
	Annual		15	**16**	**17**	**16**	**16**	**17**
NO_2	1-hr	19th Highest	200	135	148	122	130	125

Pollutants	Averaging Time		AQOs ($\mu g/m^3$)	2020	2021	2022	2023	2024
	24-hr	10th Highest	120	64	78	68	67	66
	Annual		40	32	40	37	37	35
SO_2	10-min	4th Highest	500	26	24	21	20	14
	24-hr	4th Highest	40	10	14	7	10	5

Notes:
[1] Monitoring results exceeded AQOs are shown as bold and underlined characters.
[2] All data is extracted from the Smart Air Monitoring Platform (SAMP) v2.1 (https://aqia.epd.gov.hk/)
[3] Reference conditions of gaseous pollutants concentration data: 293K and 101.325 kPa.

Table 3.4 Average Concentrations of Pollutants in the Recent Five Years (Year 2020 – 2024) at North Air Quality Monitoring Station

Pollutants	Averaging Time		AQOs (µg/m3)	2020	2021	2022	2023	2024
RSP	24-hr	10th Highest	75	55	62	50	57	56
	Annual		30	-	25	23	27	25
FSP	24-hr	19th Highest	37.5	29	29	28	28	29
	Annual		15	-	15	14	15	**16**
NO_2	1-hr	19th Highest	200	112	135	115	116	114
	24-hr	10th Highest	120	54	70	58	58	56
	Annual		40	-	36	31	30	29
SO_2	10-min	4th Highest	500	19	18	27	27	18
	24-hr	4th Highest	40	8	7	7	7	7

Notes:
[1] Since the North air quality monitoring station was commissioned on 10 July 2020, there was no data for the annual concentrations in Year 2020.
[2] Monitoring results exceeded AQOs are shown as bold and underlined characters.
[3] All data is extracted from the Smart Air Monitoring Platform (SAMP) v2.1 (https://aqia.epd.gov.hk/)
[4] Reference conditions of gaseous pollutants concentration data: 293K and 101.325 kPa.

3.3.1.4 Apart from the air quality monitoring data, EPD has released a set of background levels from "Pollutants in the Atmosphere and their Transport over Hong Kong version 3.0" (PATH v3.0) model in January 2024. The PATH v3.0 model is a regional air quality model developed by EPD to simulate future background concentrations in Hong Kong. The ST Section covers 26 grids in the PATH v3.0 model which are shown in **Figure 3.2**. Considering that the ST Section is tentatively scheduled to commence operation in 2036, the most recent data available PATH v3.0 is Year 2035, as shown in **Table 3.5**. The predicted background concentrations of NO_2, SO_2, RSP, and FSP at various averaging times are all well below the respective AQOs.

Table 3.5 Predicted Background Air Pollutants Concentration in Year 2035 (Extracted from the SAMP v2.1)

Pollutants		RSP		FSP		NO_2			SO_2	
Averaging Time		24-hr 10th Highest	Annual	24-hr 19th Highest	Annual	1-hr 19th Highest	24-hr 10th Highest	Annual	10-min 4th Highest	24-hr 4th Highest
AQOs (μg/m³)		75	30	37.5	15	200	120	40	500	40
PATH Grid Cell	(27,47)	52	20	31	12	61	25	13	24	6
	(27,48)	54	21	32	13	62	24	12	24	6
	(27,49)	51	20	29	12	64	23	12	24	7
	(28,46)	52	20	31	12	57	21	11	24	6
	(28,47)	52	20	31	12	57	22	11	23	6
	(28,48)	54	21	32	13	60	21	11	24	6
	(28,49)	50	20	29	12	57	21	11	25	7
	(28,50)	54	21	31	13	63	22	12	24	7
	(29,49)	50	19	30	12	53	18	9	25	6
	(29,50)	54	21	32	13	57	20	11	24	7
	(30,49)	51	20	30	12	47	17	9	25	6
	(30,50)	52	20	32	13	51	18	10	25	7
	(30,51)	51	20	31	12	54	19	10	25	7
	(31,50)	52	20	30	12	46	16	9	26	6
	(31,51)	51	20	30	12	50	17	9	26	7
	(31,52)	54	21	31	13	56	20	11	25	7
	(31,53)	53	20	31	13	63	21	12	26	7
	(31,54)	54	21	31	13	66	22	12	26	7
	(32,51)	52	20	31	13	50	17	10	29	7
	(32,52)	55	21	32	13	55	19	11	25	7
	(32,53)	54	21	32	13	61	21	12	26	7
	(32,54)	54	21	32	13	67	21	12	26	7
	(33,51)	54	21	33	13	48	17	10	27	7
	(33,52)	56	21	34	13	53	19	11	26	7
	(33,53)	55	21	33	13	60	21	12	26	7
	(33,54)	55	21	32	13	66	21	12	27	7

3.4 Identification of Representative Air Sensitive Receiver

3.4.1.1 In accordance with Annex 12 of the EIAO-TM, any domestic premises, hotel, hostel, hospital, clinic, nursery, temporary housing accommodation, school, educational institution, office, factory, shop, shopping centre, place of public worship, library, court of law, sport stadium or performing arts centre are considered as air sensitive receivers (ASRs). Any other premises or place with which, in terms of duration or number of people affected, has a similar sensitivity to the air pollutants as the aforelisted premises and places shall also be considered to be a sensitive receiver.

3.4.1.2 In accordance with Clause 3.5.3.2 of the EIA Study Brief (ESB-373/2025), the study area for air quality impact assessment is defined as a distance of 500m from the boundary of the ST Section and its associated works (hereinafter as “the 500m assessment area”). The 500m assessment area is shown in **Figure 3.1.**

3.4.1.3 For identification of the representative ASRs within the assessment area that would likely be affected by the potential impacts from the construction and operation of the Project, a review has been conducted based on relevant available information including topographic maps, Outline Zoning Plans (OZPs) and other published plans (including Development Permission Area Plans, Outline Development Plans, Layout Plans and land use and development applications approved by the Town Planning Board) in the vicinity of the Project site. The representative ASRs within the assessment area are presented in **Table 3.6** and illustrated in **Figures 3.2**, **3.2.1**, **3.2.2**, **3.2.3**, and **3.2.4**. It is confirmed that the selected ASRs listed in **Table 3.6** are representative due to their proximity to the Project for this assessment, based on the best available information at the time of the EIA report.

Table 3.6 Representative Air Sensitive Receivers

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
Existing ASRs						
A001	Au Tau Youth Centre, 2C Castle Peak Road - Yuen Long	G/IC	272	1.5 - 5	✓	✓
A002	HK Boy Scouts Association Yuen Long East District Centre	G/IC	281	1.5 - 10	✓	✓
A003	Harmonic Villa Block A, 2 Yau Shin Street	R	462	1.5 - 10	✓	✓
A004	Shin Leung House, Long Shin Estate, 11 Yau Shin Street	R	493	1.5 - 60	✓	✓
A005	TWGHs C Y Ma Memorial College, 3 Yau Shin Street	EDU	402	1.5 - 10	✓	✓
A006	Temporary Structure near Tung Shung Lei	R	19	1.5 - 5	✓	✓
A007	29 Tung Shing Lei	R	345	1.5 - 5	✓	✓
A008	32 Tung Shing Lei	R	357	1.5 - 5	✓	✓
A009	33 Tung Shing Lei	R	360	1.5 - 5	✓	✓
A010	37 Tung Shing Lei	R	360	1.5 - 10	✓	✓
A011	The Church of Jesus Christ of Latter-Day Saints, 1 Tung Shing Lei	WOR	367	1.5 - 10	✓	✓
A012	63 Tung Shing Lei	R	251	1.5 - 10	✓	✓
A013	Joyful House at DD103 1000	R	282	1.5 - 5	✓	✓
A014	Little Club House at DD103 346	R	278	1.5 - 5	✓	✓
A015	Anchors Academy Affiliated International Kindergarten cum Nursery	EDU	23	1.5 - 25	✓	✓
A016	Football Field at Anchors Academy Affiliated International Kindergarten cum Nursery	EDU	11	1.5	✓	✓
A017	AFCD Au Tau Fisheries Office	O	203	1.5 - 5	✓	✓
A018	Crescent Green Tower 6	R	23	1.5 - 50	✓	✓
A019	Crescent Green Tower 10	R	28	1.5 - 50	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
A020	3A Ha Ko Po Tsuen	R	72	1.5 - 10	✓	✓
A021	Village House in DD103 281 S.A ss.2 S.D in Ha Ko Po Tsuen	R	102	1.5 - 10	✓	✓
A022	Park Yoho Venezia Tower 7A	R	120	1.5 - 35	✓	✓
A023	Park Yoho Tower 8A, 18 Castle Peak Road - Tam Mi	R	333	1.5 - 45	✓	✓
A024	York International Kindergarten	EDU	188	1.5 - 10	✓	✓
A025	Riva Helorus Boulevard Tower 2	R	105	1.5 - 70	✓	✓
A026	Riva Helorus Boulevard Tower 3	R	85	1.5 - 70	✓	✓
A027	Riva Genova Boulevard Block 5	R	148	1.5 - 45	✓	✓
A028	Genova Boulevard Block 1, Riva, 1 Ying Ho Road	R	181	1.5 - 20	✓	✓
A029	Cervia Boulevard House 1, Riva, 1 Ying Ho Road	R	211	1.5 - 10	✓	✓
A030	Ferentino Boulevard Block 1, Riva, 1 Ying Ho Road	R	209	1.5 - 20	✓	✓
A031	Bevera Boulevard House 1, Riva, 1 Ying Ho Road	R	241	1.5 - 15	✓	✓
A032	Enna Boulevard Block 5, Riva, 1 Ying Ho Road	R	248	1.5 - 45	✓	✓
A033	Avio Boulevard House 1, Riva, 1 Ying Ho Road	R	313	1.5 - 15	✓	✓
A034	307 Fung Kat Heung	R	62	1.5 - 5	✓	✓
A035	Fung Kat Heung Basketball Court at GLA-YL 547	G/IC	310	1.5	✓	✓
A036	Yue Cheong Yuen at DD107 569 near Fung Kat Heung Road	R	402	1.5 - 5	✓	✓
A037[9]	Construction Sector Imported Labour Quarters D1	R	38	1.5 - 5	✓	
A038[10]	Koon Chun Hing Kee Soy & Sauce Factory Ltd	O	6	1.5 - 5	✓	
A039	15 Mo Fan Heung	R	5	1.5 - 5	✓	✓
A040	19 Mo Fan Heung	R	2	1.5 - 5	✓	✓
A041	79A Mo Fan Heung	R	38	1.5 - 5	✓	✓
A042	Park Yoho Sicilia Tower 1A	R	105	1.5 - 45	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
A043	Park Yoho Venezia Tower 2B	R	141	1.5 - 45	✓	✓
A044	Park Vista 1A Tower 11B	R	253	1.5 - 50	✓	✓
A045	Park Yoho Genova Tower 17	R	345	1.5 - 50	✓	✓
A046	Park Yoho Napoli Tower 28	R	442	1.5 - 40	✓	✓
A047	Park Yoho Milano Tower 31A	R	434	1.5 - 35	✓	✓
A048	7 Sha Po Tsuen	R	312	1.5 - 10	✓	✓
A049	Basketball Court at 400 Sha Po Tsuen	G/IC	274	1.5	✓	✓
A050	Construction Sector Imported Labour Quarters S1	R	59	1.5 - 5	✓	✓
A051	Temporary Structure in DD104 4202 RP	R	21	1.5 - 5	✓	✓
A052	106 Mo Fan Heung	R	60	1.5 - 5	✓	✓
A053	171 Mo Fan Heung	R	46	1.5 - 5	✓	✓
A054	Merry Garden House 1	R	24	1.5 - 10	✓	✓
A055	166 Pok Wai	R	Within site boundary	1.5 - 10	✓	✓
A056	169 Pok Wai	R	Within site boundary	1.5 - 10	✓	✓
A057	170 Pok Wai	R	Within site boundary	1.5 - 10	✓	✓
A058	172 Pok Wai	R	Within site boundary	1.5 - 5	✓	✓
A059	177 Pok Wai	R	Within site boundary	1.5 - 10	✓	✓
A060	181 Pok Wai	R	Within site boundary	1.5 - 10	✓	✓
A061	230 Pok Wai	R	Within site boundary	1.5 - 10	✓	✓
A062	119B Pok Wai (Grenville)	R	113	1.5 - 10	✓	✓
A063	138B-138 Pok Wai, Elite Garden	R	36	1.5 - 10	✓	✓
A064	138J Pok Wai, Elite Garden	R	22	1.5 - 15	✓	✓
A065	Village House in DD104 3722 S.B ss.6 S.D near Wing Kei Tsuen	R	114	1.5 - 10	✓	✓
A066	6E Wing Kei Tsuen	R	117	1.5 - 15	✓	✓
A067	Greenery Gardens 23, 3 Fairview Park Boulevard	R	397	1.5 - 10	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
A068	Meister House Tower 1, 1 Fairview Park Boulevard	R	312	1.5 - 10	✓	✓
A069	Village House in DD104 1895 S.B RP	R	159	1.5 - 5	✓	✓
A070	Village House in DD104 1831 near San Wai Tsuen	R	160	1.5 - 5	✓	✓
A071	Village House in DD104 1976 S.A near San Wai Tsuen	R	160	1.5 - 10	✓	✓
A072	221 San Wai Tsuen	R	246	1.5 - 10	✓	✓
A073	Wah On Villa Block 25, 138 San Wai Tsuen	R	251	1.5 - 10	✓	✓
A074	Ian Court Block 21	R	342	1.5 - 10	✓	✓
A075	Temporary Structure to the Southeast of Lamp Post VD3741 near Ching Yau Road	R	Within site boundary	1.5 - 5	✓	✓
A076	Temporary Structure (village house) at Ngau Tam Mei	R	Within Site Boundary	1.5 - 5	✓	✓
A077	Office at Ngau Tam Mei Animal Waste Composting Plant	O	Within Site Boundary	1.5 - 5	✓	✓
A078	Office at Ngau Tam Mei Fresh water Primary Service Reservoir	O	Within Site Boundary	1.5 - 10	✓	✓
A079	Temporary Structure (village house) near Lamp Post VD5446	R	Within Site Boundary	1.5 - 5	✓	✓
A080	Village House to the Southwest of Lamp Post VD2284 near Ching Yau Road	R	Within Site Boundary	1.5 - 5	✓	✓
A081	Temporary Structure (village house) at Ngau Tam Mei	R	Within site boundary	1.5 - 5	✓	✓
A082	337 San Wai Tsuen	R	72	1.5 - 10	✓	✓
A083	63B San Wai Tsuen	R	319	1.5 - 10	✓	✓
A084	Temporary Structure (village house) near Lamp Post VD1836	R	Within site boundary	1.5 - 5	✓	✓
A085	Village House	R	Within site boundary	1.5 - 5	✓	✓
A086	House at Lot No.163 in D.D.104	R	Within site boundary	1.5 - 5	✓	✓
A087	Yiu Shing Mo Temple	WOR	Within site boundary	1.5 - 10	✓	✓
A088	Ngau Tam Mei Water Treatment Works Administration Building	O	102	1.5 - 10	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
A089	San Tin Barracks Block 77	B	353	1.5 - 5	✓	✓
A090	291 Chau Tau Tsuen	R	402	1.5 - 10	✓	✓
A091	Village House at Ki Lun Tsuen South	R	35	1.5 - 5	✓	✓
A092	Village House to the Southwest of Lamp Post VD6740 in Cheung Lek	R	47	1.5 - 5	✓	✓
A093	Village House to the Southwest of Lamp Post H3353 near Ki Lun Tsuen	R	33	1.5 - 10	✓	✓
A094	Village House to the Southeast of Lamp Post H3354 near Ki Lun Tsuen	R	96	1.5 - 10	✓	✓
A095	Village House to the South of Lamp Post VD0256 near Ki Lun Tsuen	R	129	1.5 - 10	✓	✓
A096	Village House to the South of Lamp Post VD0260 near Ki Lun Tsuen	R	56	1.5 - 10	✓	✓
A097	Village House to the Northwest of Lamp Post FA4049 near Ki Lun Tsuen	R	23	1.5 - 10	✓	✓
A098	Village type house to the north of Ki Lun Tsuen Playground	R	Within site boundary	1.5 - 5	✓	✓
A099	Village type house to the north of Ki Lun Tsuen Playground	R	Within site boundary	1.5 - 10	✓	✓
A100	Village type house to the south of Ki Lun Tsuen Playground	R	Within site boundary	1.5 - 5	✓	✓
A101	Village type house to the south of Ki Lun Tsuen Playground	R	Within site boundary	1.5 - 5	✓	✓
A102	Europa Garden Block B6	R	Within site boundary	1.5 - 15	✓	✓
A103	Europa Garden Block B21	R	Within site boundary	1.5 - 15	✓	✓
A104	Europa Garden Block B36	R	Within site boundary	1.5 - 15	✓	✓
A105	Europa Garden Block A1	R	Within site boundary	1.5 - 15	✓	✓
A106	Europa Garden Block A15	R	37	1.5 - 15	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
A107	Europa Garden Block A27	R	69	1.5 - 10	✓	✓
A108	Kwu Tung Market Shopping Centre	G/IC	52	1.5 - 15	✓	✓
A109	Lady Ho Tung Welfare Centre Eco-Learn Institute	G/IC	107	1.5 - 15	✓	✓
A110	10 Kwu Tung South Road	R	281	1.5 - 5	✓	✓
A111	31 Rigi Avenue, Valais	R	121	1.5 - 10	✓	✓
A112	9 Neuchatel Avenue, Valais	R	170	1.5 - 10	✓	✓
A113	21 Neuchatel Avenue, Valais	R	94	1.5 - 10	✓	✓
A114	21 Locarno Avenue, Valais	R	192	1.5 - 10	✓	✓
A115	12 Locarno Avenue, Valais	R	167	1.5 - 10	✓	✓
A116	Village type house at Ma Tso Lung	R	198	1.5 - 10	✓	✓
A117	Village type house at Tit Hang	R	422	1.5 - 5	✓	✓
A118	Ngai's Temple, 19 Cheung Lek	WOR	416	1.5 - 5	✓	✓
A119	15A Cheung Lek	R	406	1.5 - 10	✓	✓
A120	561 Hang Tau	R	482	1.5 - 10	✓	✓
A121	Grand Garden Block 5, 6a Hang Tau	R	485	1.5 - 15	✓	✓
A122	412 Hang Tau	R	446	1.5 - 10	✓	✓
A123	26 Gallen Boulevard, Valais	R	341	1.5 - 10	✓	✓
A124	9 Moritz Avenue, Valais	R	332	1.5 - 10	✓	✓
A125	17 Geneva East Avenue, Valais	R	340	1.5 - 10	✓	✓
A126	27 Geneva South Avenue, Valais	R	337	1.5 - 10	✓	✓
A127	Shum Residence	WOR	109	1.5 - 10	✓	✓
A128	Chan Ming Yam Tong	WOR	50	1.5 - 5	✓	✓
A129	Pok Oi Hospital	G/IC	495	1.5 - 60	✓	✓
A130	Village Houses in D.D. 104	R	Within site boundary	1.5 - 5	✓	✓
A131	Village House in D.D. 100	R	Within site boundary	1.5 - 5	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
A132	Kwu Tung North Multi-welfare Services Complex	G/IC	54	1.5 - 30	✓	✓
A133	Village house at Kam Tin	R	166	1.5 - 10	✓	✓
A134	Temporary Structure (shop) at Nam Sang Wai	C	16	1.5	✓	✓
A135	Temporary Structure (office) near Fung Kat Heung	O	19	1.5 - 5	✓	✓
A136	Village house at Mo Fan Heung	R	Within Site Boundary	1.5 - 5	✓	✓
A137	Camp Site of Forest Pathway	REC	55	1.5	✓	✓
A138	Sun Hing Farm	C	Within Site Boundary	1.5	✓	✓
A139	Camp Site of Aqualand Camping	REC	214	1.5	✓	✓
A140	Lohas Organic Farm	C	393	1.5	✓	✓
Planned ASRs						
PA001[3]	Proposed Public Housing Development at Sha Po	R	108	1.5 - 185	✓	✓
PA002[3]	Proposed Government, Institution or Community Facility	G/IC	52	1.5 - 185	✓	✓
PA003[3]	Proposed Public Housing Development at Sha Po	R	21	1.5 - 185	✓	✓
PA004[4]	Proposed Government, Institution or Community Facility	G/IC	Within Site Boundary	1.5 - 40	✓	✓
PA005[4]	Proposed Private Residential atop Ngau Tam Mei Depot	R	16	1.5 - 210	✓	✓
PA006[4]	Proposed Fire Station cum Ambulance Depot and Staff Quarters	G/IC	6	1.5 - 120	✓	✓
PA007[4]	Proposed Private Residential	R	83	1.5 - 165	✓	✓
PA008[4]	Proposed Private Residential	R	226	1.5 - 170	✓	✓
PA009[4]	Proposed Hospital	G/IC	20	1.5 - 125	✓	✓
PA010[4]	Proposed Hospital	G/IC	24	1.5 - 125	✓	✓
PA011[4]	Proposed Government, Institution or Community Facility	EDU	268	1.5 - 40	✓	✓
PA012[4]	Proposed University	EDU	4	1.5 - 130	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
PA013[4]	Proposed University	EDU	20	1.5 - 125	✓	✓
PA014[4]	Proposed University	EDU	Within site boundary	1.5 - 125	✓	✓
PA015[4]	Proposed University	EDU	Within site boundary	1.5 - 125	✓	✓
PA016[4]	Proposed University	EDU	Within site boundary	1.5 - 120	✓	✓
PA017[5]	Planned Information and Technology - Zone 4	I&T	Within site boundary	1.5 - 120	✓	✓
PA018[5]	Planned Logistics, Storage and Warehouse	LSW	Within site boundary	1.5 - 150	✓	✓
PA019[5]	Planned Information and Technology - Zone 4	I&T	Within site boundary	1.5 - 130	✓	✓
PA020[5]	Planned Logistics, Storage and Warehouse	LSW	176	1.5 - 150	✓	✓
PA021[5]	Planned Information and Technology - Zone 4	I&T	337	1.5 - 140	✓	✓
PA022[5]	Planned Information and Technology - Zone 4	I&T	499	1.5 - 90	✓	✓
PA023[5]	Planned Information and Technology - Zone 4	I&T	214	1.5 - 140	✓	✓
PA024[5]	Planned Information and Technology - Zone 4	I&T	430	1.5 - 130	✓	✓
PA025[5]	Planned Logistics, Storage and Warehouse	LSW	187	1.5 - 130	✓	✓
PA026[5]	Planned Information and Technology - Zone 3	I&T	364	1.5 - 100	✓	✓
PA027[5]	Planned Information and Technology - Zone 3	I&T	386	1.5 - 100	✓	✓
PA028[5]	Planned Information and Technology - Zone 3	I&T	222	1.5 - 90	✓	✓
PA029[5]	Planned Information and Technology - Zone 3	I&T	157	20 - 70	✓	✓
PA030[5]	Planned Information and Technology - Zone 3	I&T	111	20 - 70	✓	✓
PA031[5]	Planned Information and Technology - Zone 4	I&T	Within site boundary	1.5 - 120	✓	✓
PA032[6]	Proposed Government, Institution or Community Facility	G/IC	64	1.5 - 120	✓	✓
PA033[6]	Proposed Government, Institution or Community Facility	G/IC	5	1.5 - 120	✓	✓
PA034[6]	Proposed Secondary School	EDU	26	1.5 - 25	✓	✓
PA035[6]	Proposed Hospital, Polyclinic and General Clinic / Health Centre	G/IC	Within site boundary	1.5 - 115	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
PA036[6]	Proposed Hospital, Polyclinic and General Clinic / Health Centre	G/IC	33	1.5 - 115	✓	✓
PA037[6]	Proposed Government, Institution or Community Facility	G/IC	8	1.5 - 115	✓	✓
PA038[6]	Proposed Residential Building	R	452	1.5 - 100	✓	✓
PA039[6]	Proposed Residential Building	R	446	1.5 - 100	✓	✓
PA040[6]	Proposed Residential Building	R	405	1.5 - 120	✓	✓
PA041[6]	Proposed Residential Building	R	364	1.5 - 120	✓	✓
PA042[6]	Proposed Residential Building	R	220	1.5 - 110	✓	✓
PA043[6]	Proposed Primary School	G/IC	479	1.5 - 25	✓	✓
PA044[6]	Proposed Government, Institution or Community Facility	G/IC	448	1.5 - 100	✓	✓
PA045[6]	Proposed Government, Institution or Community Facility	G/IC	489	1.5 - 100	✓	✓
PA046[6]	Proposed Business and Technology Park	C	338	1.5 - 50	✓	✓
PA047[6]	Proposed Residential Building	R	268	1.5 - 115	✓	✓
PA048[6]	Proposed Residential Building	R	322	1.5 - 115	✓	✓
PA049[7]	Proposed Residential Building of Land Sharing Pilot Scheme Application No. LSPS/002	R	265	1.5 - 95	✓	✓
PA050[7]	Proposed Residential Building of Land Sharing Pilot Scheme Application No. LSPS/002	R	200	1.5 - 95	✓	✓
PA051[7]	Proposed Residential Building	R	495	1.5 - 105	✓	✓
PA052[5]	Planned Information and Technology - Zone 3	I&T	92	1.5 - 70	✓	✓
PA053[8]	Planned Residential Building in Planning Application no. Y/YL-NSW/7	R	145	1.5 - 50	✓	✓
PA054[8]	Planned Residential Building in Planning Application no. Y/YL-NSW/7	R	136	1.5 - 50	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
PA055[9]	Propose Residential Building in Planning Application no. Y/YL-NSW/8	R	Within site boundary	1.5 - 110	✓	✓
PA056[9]	Propose Residential Building in Planning Application no. Y/YL-NSW/8	R	<5	1.5 - 110	✓	✓
PA057[10]	Planned Residential Building in Planning Application no. Y/YL-NSW/9	R	<5	1.5 - 115	✓	✓
PA058[10]	Planned Residential Building in Planning Application no. Y/YL-NSW/9	R	Within site boundary	1.5 - 110	✓	✓
PA059[11]	Planned residential development under the Planning Application no. A/YL-NSW/274	R	431	1.5 - 70	✓	✓
PA060[11]	Planned commercial, kindergarten, welfare facilities under the Planning Application no. A/YL-NSW/274	C, EDU, G/IC	457	1.5 - 10	✓	✓
PA061[11]	Planned residential development under the Planning Application no. A/YL-NSW/274	R	391	1.5 - 70	✓	✓
PA062[11]	Planned residential development under the Planning Application no. A/YL-NSW/274	R	479	1.5 - 70	✓	✓
PA063[12]	Planned residential development under the Planning Application no. A/YL-NSW/293	R	134	1.5 - 135	✓	✓
PA064[12]	Planned residential development under the Planning Application no. A/YL-NSW/293	R	81	1.5 - 135	✓	✓
PA065[12]	Planned residential development under the Planning Application no. A/YL-NSW/293	R	54	1.5 - 15	✓	✓
PA066[12]	Planned residential development under the Planning Application no. A/YL-NSW/293	R	207	1.5 - 135	✓	✓
PA067[12]	Planned residential development under the Planning Application no. A/YL-NSW/293	R	137	1.5 - 135	✓	✓
PA068[13]	Planned residential development under the Planning Application no. A/YL-NSW/314	R	260	1.5 - 15	✓	✓

ID	Description	Land Use[1]	The Shortest Distance between the ASR and the ST Section Project Boundary (m)	Assessment Height Above Ground (mAG)[2]	Potentially Affected during Construction Phase	Potentially Affected during Operation Phase
PA069[13]	Planned residential development under the Planning Application no. A/YL-NSW/314	R	361	1.5 - 15	✓	✓
PA070[13]	Planned residential development under the Planning Application no. A/YL-NSW/314	R	429	1.5 - 15	✓	✓
PA071[10]	Planned School in Planning Application no. Y/YL-NSW/9	EDU	Within Site Boundary	1.5 - 20	✓	✓
PA072[10]	Planned Factory in Planning Application no. Y/YL-NSW/9	I	Within Site Boundary	1.5 - 10	✓	✓
PA073	Planned NMH Administration Building	O	Within Site Boundary	1.5 - 30		✓
PA074[14]	Planned residential development under the Planning Application no. A/YL-KTN/964	R	153	1.5 - 60	✓	✓

Notes:

[1] R – Residential; EDU – Education; G/IC – Government, Institution and Community; WOR – Place of Public Worship; O – Office; C – Commercial; B – Barrack; LSW – Logistic Storage and Warehouse; I&T – Information and Technology; REC – Recreational; I – Industrial;

[2] Full range of assessment height for each ASR has been considered.

[3] There is no information available regarding Sha Po Public Residential Development during this EIA Study. According to the District Council Discussion Papers 2020-2023 Document No. 36/2022, titled “Proposed amendment to Kam Tin North Outline Zoning Plan (Approved Plan No. S/YL-KTN/9) and public housing development at Sha Po, Yuen Long”, the maximum height of proposed public housing development is 185mAG. Thus, the maximum height of proposed public housing development is adopted for PA001 to PA003.

[4] The locations and height of planned representative ASRs at Ngau Tam Mei Development are based on the approved EIA report Development at Ngau Tam Mei Area (AEIAR-271/2025). The completion year of the proposed land uses will be Year 2033.

[5] The locations, heights, and land uses of the planned representative ASRs at the San Tin/Lok Ma Chau Development Node are detailed in the approved Environmental Impact Assessment (EIA) study for the San Tin/Lok Ma Chau Development Node (AEIAR-261/2024).

[6] For the Kwu Tung North New Development Area (NDA), the assessment of heights and land uses for the planned representative ASRs is based on the "Environmental Review Report of Agreement No. CE 19/2019(CE) for the Development of Kwu Tung North New Development Area, Remaining Phase – Design and Construction”.

[7] The locations, the proposed land uses and maximum heights of the planned representative ASRs at Nam Sang Wai are referred to the amendments to Land Sharing Pilot Scheme Application No. LSPS/002. The completion year of the proposed land uses will be Year 2031.

[8] The locations, the proposed land uses and maximum heights of the planned representative ASRs at Nam Sang Wai are referred to the Planning Application Y/YL-NSW/7. The completion year of the proposed land uses will be Year 2030.

[9] The locations, the proposed land uses and maximum heights of the planned representative ASRs at Nam Sang Wai are referred to the Planning Application Y/YL-NSW/8. The completion year of the proposed land uses will be Year 2031. A037 will only be existed in or before Year 2031.

[10] The locations, the proposed land uses and maximum heights of the planned representative ASRs at Nam Sang Wai are referred to the Planning Application Y/YL-NSW/9. The completion year of the proposed land uses will be Year 2031. A038 will only be existed in or before Year 2031.

[11] The locations, proposed land use, and maximum height of the planned development were referenced to the approved Planning Application No. A/YL-NSW/274. No publicly available information on this development's construction and occupation programme.

[12] The locations, proposed land use, and maximum height of the planned development were referenced to the approved Planning Application No. A/YL-NSW/293. No publicly available information on this development's construction and occupation programme.

[13] The locations, proposed land use, and maximum height of the planned development were referenced to the approved Planning Application No. A/YL-NSW/314. No publicly available information on this development's construction and occupation programme.

[14] The locations, proposed land use, and maximum height of the planned development were referenced to the approved Planning Application No. A/YL-KTN/964. No publicly available information on this development's construction and occupation programme.

3.5 Identification of Environmental Impact

3.5.1 Construction Phase

3.5.1.1 The construction air quality impact assessment would be evaluated qualitatively in accordance with Clause 1.2 of Appendix B of the EIA Study Brief and Annex 12 of the EIAO-TM.

3.5.1.2 As discussed in **Section 2.7**, the Project includes site formation and excavation work for the construction of at-grade and viaduct roads, tunnel construction by drill-and-blast, construction of administration buildings and satellite control buildings, and slope works, etc. Construction of the Project is anticipated to commence tentatively in Year 2027 with a target to commission in or before Year 2036. The detailed construction program is presented in **Section 2.8** and **Table 2.4**.

3.5.1.3 The construction works would likely generate fugitive dust and gaseous emission with potential impacts on neighbouring ASRs from various construction activities, including site formation, site clearance, excavation, slope works, construction of administration buildings and satellite control buildings, blasting for tunnel construction, mucking out blasted rock and soil from drill-and-blast operations, material handling and backfilling, foundation works, handling and stockpiling of dusty materials, wind erosion of exposed area, use of on-site powered mechanical equipment, and emissions from construction vehicles, etc. The locations of these dusty works areas are presented in **Appendix 3.1**.

3.5.1.4 Vehicular emissions from induced construction traffic for the transportation of spoils/excavated materials would cause potential gaseous and particulates emission. On-site use of diesel-powered engines is also the potential source for gaseous pollutants, such as NOx, SO_2, CO and smoke.

3.5.2 Concurrent Projects

3.5.2.1 The status of concurrent projects within the 500m assessment area of the ST Section is summarized in **Section 2.9** and **Figure 2.3**. Cumulative construction air quality impact is anticipated for the projects shown in **Table 3.7**.

Table 3.7 Projects with Cumulative Construction Air Quality Impact

No.	Project	Tentative construction year
1	Site Formation and Infrastructure Works for Public Housing Development at Sha Po, Shap Pat Heung and Tai Kei Leng Yuen Long	Under review by the Government
2	Ngau Tam Mei New Development Area	Completion by 2036
3	San Tin Technopole	Completion by 2039
4	Kwu Tung North New Development Area	Completion by 2031
5	Northern Link	Completion by 2034
6	Ngau Tam Mei Water Treatment Works Extension	Completion by 2030
7	Proposed Nam Sang Wai Wetland Conservation Park	Under review by the Government
8	Planned residential development under Land Sharing Pilot Scheme Application No. LSPS/002	Completion in 2031
9	NMH – Tin Shui Wai Section	3-4 years after NMH-San Tin Section completion

No.	Project	Tentative construction year
10	NMH – Kwu Tung Section	3-4 years after NMH-San Tin Section completion
11	Planned residential development under Planning Application no. Y/YL-NSW/7	Completion in 2030
12	Planned residential development under Planning Application no. Y/YL-NSW/8	Completion in 2031
13	Planned residential development under Planning Application no. Y/YL-NSW/9	Completion in 2031
14	Planned residential development under Planning Application no. Y/YL-NSW/274	No publicly available information. It is considered in cumulative construction air quality impact as a conservative approach
15	Planned Residential Development under Planning Application no. A/YL-NSW/314	No publicly available information. It is considered in cumulative construction air quality impact as a conservative approach
16	Planned Residential Development Under the Planning Application no. A/YL-NSW/293	No publicly available information. It is considered in cumulative construction air quality impact as a conservative approach
17	Planned Residential Development Under the Planning Application no. A/YL-KTN/964	No publicly available information. It is considered in cumulative construction air quality impact as a conservative approach
18	Planned Residential Development under the Planning Application no. A/YL-KTN/604 and A/YL-KTN/663	No publicly available information. It is considered in cumulative construction air quality impact as a conservative approach
19	Planned Social Welfare Facilities under the Planning Application no. A/YL-NSW-348 and A/YL-NSW-349	No publicly available information. It is considered in cumulative construction air quality impact as a conservative approach
20	Planned School under the Planning Application no. Y/NE-KTS/18	No publicly available information. It is considered in cumulative construction air quality

No.	Project	Tentative construction year
		impact as a conservative approach

3.5.3 Operation Phase

3.5.3.1 During the Project planning stage, due consideration has been given to minimise the air pollution impact on the nearby ASRs. Specifically, the locations of the tunnel portal exits at Hadden Hill have been situated away from the ASRs as far as practicable to minimise air pollution impact associated with emissions from the tunnel portals. In addition, the alignment of the aboveground sections near Long Ha and Ngau Tam Mei have been carefully designed to transverse along the hillside, thereby maximising the separation distance from the existing ASRs at Long Ha and the planned ASRs at 'Development of Ngau Tam Mei' (NTM Project). These planning considerations aim to reduce potential operational air quality impacts on the ASRs.

3.5.3.2 Potential air quality impacts during the operation phase of the ST Section would be associated with the following pollution sources.

Within the 500m assessment area:

- Background pollutant concentrations;
- Vehicular emission from open sections of existing, planned, and proposed road networks within 500m assessment area;
- Emission from the existing underpass portal near Koon Chun Sauce Factory and proposed tunnel portals of the ST Section;
- Industrial emissions; and
- Emissions associated with the existing and planned public transport interchange (PTI), bus and minibus termini, and parking sites;

Within the 4km of the ST Section Project Boundary***:***

- Major point source from K. Wah Asphalt Ltd

Vehicular emissions from the Proposed and Other Roads

3.5.3.3 Vehicular emission from the proposed and other roads including Tsing Long Highway, Yuen Long Highway, Castle Peak Road – Yuen Long, Kam Tin Road, San Tin Highway, San Tam Road, Castle Peak Road – Tam Mi, and Fanling Highway, etc. within 500m assessment area contributes to the ambient air quality. These emissions include open road emission and portal emission from the tunnel sections of the ST Section and an underpass near Koon Chun Sauce Factory. The locations of portal emission are shown in **Figure 3.3**.

Emissions associated with the existing and planned public transport interchange (PTI), bus and minibus termini, and heavy goods vehicles and Coach parking sites

3.5.3.4 Existing and planned public transport interchange (PTI), bus and minibus termini, and heavy goods vehicles (HGV) and coach parking sites with frequent operation associated with vehicles also contribute to ambient air quality by vehicular running, idling, and start emission within these facilities. Cold start emission surveys were conducted continuously for 24 hours on a normal working day for all identified facilities. The HGV and coach parking sites, bus and minibus termini, and PTI within the 500m assessment area were identified and presented in **Table 3.8**, and their locations are illustrated in **Figure 3.4**. The emissions from these concerned facilities would be included in the modelling assessment. No on-street parking of HGV and Non-Franchised Buses (NFB) have been identified within the assessment area.

Table 3.8 Existing and Planned Public Transport Interchange (PTI), Bus and Minibus Termini, and Heavy Goods Vehicles (HGV) and Coach Parking Sites

ID	Description
1	Planned PTI at Ngau Tam Mei
2	Park YOHO PTI
10	Kim Logistics Co. Ltd.
13	Warehouse / vehicle repair workshops along Pok Wai West Road
14	C-H Logistic
15	Hichain Logistics Company Limited
18	Kwu Tung North Multi-Welfare Services Complex Terminus
19	Planned PTI at the private residential development under the planning application nos. Y/YL-NSW/8 and Y/YL-NSW/9
20	Planned PTI at Public Housing Development at Sha Po
21	Planned PTI at Kwu Tung North
22	Planned Transport Interchange Hub at San Tin / Lok Ma Chau Development Node
24	Planned PTI at Planned Residential Development under the Planning Application No. A/YL-KTN/604

3.5.3.5 Based on the site survey findings, some of the existing termini / parking sites were identified with only limited or no cold start emission, including infrequent trips, no start emission generated, no HGV or NFB parking, etc. These termini / parking sites have therefore been excluded from the assessment, as listed in **Table 3.9** and shown in **Appendix 3.5**.

Table 3.9 Summary of Termini / Parking Sites Excluded in Cold Start Emission Calculation

ID	Description	Reason
3	Riva Red Public Light Bus Stop	Public Light Bus (PLB) idles without switching off the engines, as there is no proper area for parking. No start emission was generated.
4	Ngau Tam Mei West Green Public Light Bus Stop	PLB leaves after passengers drop on/off, as there is no proper area for parking. No start emission would be generated. It would also be resumed for Development at Ngau Tam Mei Area.
5	Ki Lun Tsuen Green Public Light Bus Stop	PLB leaves after passengers drop on/off, as there is no proper area for parking. No start emission would be generated.
6	Kwu Tung South Green Public Light Bus Stop	PLB idles without switching off the engine, as there is no proper area for parking. No start emission was generated.

ID	Description	Reason
7	Carpark at Au Tau	Infrequent trips (max. 5 trips per hour) were observed.
8	Carpark at Ko Po Tsuen	No HGV, NFB, PLB or FB parked.
9	Yuen Long Vegetable Union Association Limited	Infrequent trips (max. 1 trip per hour) were observed.
11	New Power Auto Services Company Limited	Infrequent trips (max. 2 trips per hour) were observed. The land would be resumed for Sha Po Public Housing Development.
12	Fung Kat Heung Logistic Center / Warehouse	The land would be resumed for Sha Po Public Housing Development.
16	Mitsubishi Fuso Depot	Infrequent trips (max. 2 trips per hour) were observed.
17	Crescent Green Red Public Light Bus Stop	PLB idles without switching off the engines, as there is no proper area for parking. No start emission was generated.
23	Carpark at San Tam Road	No HGV, NFB, PLB or FB parked.

Industrial Emissions and Major Point Sources within 4 km

3.5.3.6 The identification of industrial emission sources is based on the best available information including approved EIA reports, desktop study, in addition to the site surveys which conducted in May 2025 and August 2025. There are 4 industrial emission sources within the 500m assessment area assessed in the operational air quality impact. The industrial emission sources are listed in **Table 3.10** and their locations are illustrated in **Figure 3.5**.

Table 3.10 List of Industrial Emission Sources within the 500m Assessment Area

ID	Industrial Emission Sources
KTN	Proposed hospital at Kwu Tung North New Development Area
AWTP	Ngau Tam Mei Animal Waste Treatment Plant
LBF	Lard Boiling Facility
HP	Proposed hospital at Ngau Tam Mei New Development Area

3.5.3.7 According to the approved Planning Application no. Y/YL-NSW/9, the existing Koon Chun Sauce Factory would be relocated to within this application site, which is tentatively completed for occupation by 2031. With reference to the Environmental Assessment Report of the Planning Application, the relocated soy sauce factory would solely rely on electricity for operation instead of combustion fuels such as Towngas or diesel. No chimney stack would be installed at the relocated soy sauce factory and stack emission from the relocated soy sauce factory is not anticipated. Therefore, emissions from Koon Chun Sauce Factory are excluded from operational air quality impact assessment.

3.5.3.8 The asphalt plant at Man Kam To Road (AP1) is identified as a major point source within 4 km from the ST Section. The location of AP1 is indicated in **Figure 3.5**. The potential impact from these sources would be evaluated in the operational modelling assessment.

Identification of Key Air Pollutants during Operation Phase of the ST Section

3.5.3.9 Vehicular emission will be the dominant source of air pollutants associated with the Project. The key air pollutants associated with vehicular emission during operation phase include NOx, RSP and FSP.

Emissions from Concurrent Projects

3.5.3.10 The status of concurrent projects within the 500m assessment area of the ST Section is summarized in **Section 2.9** and **Figure 2.3**. Cumulative impact from concurrent projects such as Ngau Tam Mei New Development Area, San Tin Technopole, and Kwu Tung North New Development Area, etc. would be considered in the assessment.

3.6 Assessment Methodology

3.6.1 Construction Phase

3.6.1.1 The construction air quality impact assessment would be conducted qualitatively in accordance with Clause 1.2 of Appendix B of the EIA Study Brief (ESB-373/2025) and Annex 12 of the EIAO-TM.

3.6.1.2 A review of dust monitoring data during construction phase of similar infrastructure projects, including Tsuen Kwan O – Lam Tin Tunnel (TKO-LTT) (AEIAR-173/2013), Central Kowloon Route (CKR) (AEIAR-171/2013), Liantang/Heung Yuen Wai Boundary Control Point and Associated Works (LT/HYW BCP) (AEIAR-161/2011) and Route 16 Investigation Assignment from West Kowloon to Sha Tin - Alternative Alignment (R16) (AEIAR-022/1999). Similar to the ST Section, the TKO-LTT, CKR, LT/HYW BCP, and R16 involved the construction of at-grade/viaduct trunk roads with interchanges and tunnels constructed by drill-and-blast method.

3.6.1.3 The TKO-LTT involved construction of 4.8 km-long dual two-lane trunk road connecting Lam Tin and Tsuen Kwan O, of which 2.6 km is in the form of tunnel. At-grade interchange at Lam Tin connects the trunk road T2 and the viaduct interchange at Tseung Kwan O were constructed to connect between the trunk road T2 at Cha Kwo Ling and Cross Bay Link at Tseung Kwan O. Throughout the monitoring period between 2016 to 2024, 2 measurements of 24-hour total suspended particulate (TSP) exceeded the limit level of 260 $\mu g/m^3$. Investigation revealed that the 2 exceedance events were due to heavy vehicles travelling along the Cha Kwo Ling Road and were not related to the Project. In fact, over 99% of the measured 24-hr TSP levels at all monitoring stations were below the action level (i.e. 165 - 210 $\mu g/m^3$) during the whole construction period.

3.6.1.4 The CKR is a 4.7 km-long dual 3-lane trunk road connecting Yau Ma Tei and Kai Tak. The 3.3 km-long tunnel between Yau Ma Tei and Ma Tau Kok was constructed by drill-and-blast and cut-and-cover. At-grade/viaduct interchanges at Yau Ma Tei and Kai Tak were constructed for the connection of the existing trunk roads and local roads. Ventilation buildings and an administration building were proposed at Yau Ma Tei, Ho Man Tin, and Kai Tak. Throughout the monitoring period between 2018 to November 2025, all the measured 1-hour TSP and 24-hour TSP were below the limit level of 500 $\mu g/m^3$ and 260 $\mu g/m^3$ respectively.

3.6.1.5 Construction of an 11 km-long dual two-lane trunk road of which 5.3 km-long at-grade/viaduct with an at-grade interchange at Ping Yeung and 5.7 km-long tunnel by drill-and-blast method were carried out to connect the boundary control point at Liantang/Heung Yuen Wai and Fanling Highway. Throughout the monitoring period between 2013 to 2023, all the measured 1-hour TSP levels were below the limit level of 500 $\mu g/m^3$. There were 6 events of measurement exceeding the 1-hour TSP action level of 265 – 271 $\mu g/m^3$, 19 events of measurement exceeding the 24-hour TSP action levels of 143-156 $\mu g/m^3$ and 4 events of measurement exceeding the

24-hour TSP limit levels of 260 μg/m^3. Investigation revealed that the project minorly contributed to 2 exceedance events. The traffic dust along Lin Ma Hang Road was the major contributor to the 2 exceedance events. No further exceedance was observed with the full implementation of dust mitigation measures. As for other exceedance events, investigations revealed that all these exceedance events were not related to the project. In fact, over 99% of the 1-hour TSP and 24-hour TSP measurements were below the respective action and limit levels, and that the Final Review Environmental Monitoring & Audit Report concluded that the construction dust suppression measures as recommended in the Implementation Schedule for Environmental Mitigation Measures were proven to be effective and adequate.

3.6.1.6 The R16, later renamed as Route 8 (R8), consists of 1.4 km-long Lai Chi Kok viaduct, 0.5km of at-grade carriageway, and 2.1km-long Eagle's Nest tunnel and connects Cheung Sha Wan and Sha Tin. According to the Final Environmental Monitoring & Audit Review Reports for Lai Chi Kok viaduct and Eagle's Nest Tunnel, a total of 12 action level and 2 limit level exceedances for 1-hour TSP and 3 action level and no limit level exceedance for 24-hour TSP were recorded during the entire construction period. Upon investigation, one exceedance of 1-hour TSP action level of 294 μg/m^3 was found to be related to the excavation works and site vehicle movement from the R16 Project. No further exceedance was observed with the implementation of construction dust suppression measures. Other exceedance events were found to be related to the poor ambient air quality, traffic dust on Tai Po Road, and dust generated from other projects nearby.

3.6.1.7 In view of the insignificant dust impact caused by these past projects of similar scales, it is anticipated that the Project would not cause adverse air quality impacts during construction phase with the implementation of appropriate mitigation measures and good site practices. Therefore, qualitative assessment approach was adopted for construction dust assessment in accordance with Annex 12 of the EIAO-TM to ensure that the Air Pollution Control (Construction Dust) Regulation is complied with.

3.6.1.8 Additionally, a comprehensive EM&A programme with RSP and FSP real-time monitoring would be conducted to ensure the proper implementation of measures and the compliance of AQOs during the construction of the Project.

3.6.2 Operation Phase

3.6.2.1 PATH and AERMOD models were used to simulate the dispersion of emission from the sources identified in **Section 3.5.3**. The future background concentrations for air pollutants from the PATH v3.0 model for the Year 2035, available in the EPD Smart Air Modelling Platform v2.1 (SAMP v2.1), was adopted. AERMOD model, the EPD approved air dispersion model, was used to simulate vehicular emission associated with the open road emissions, portal emission, industrial emission, and emission associated with the existing and planned PTI, bus and minibus termini, parking sites within the 500m assessment area.

Vehicular emissions from the open sections of the existing, planned, and proposed roads

3.6.2.2 Open section of existing roads, proposed and planned road networks (including at-grade roads and viaduct at San Tin Highway, Castle Peak Road – Tam Mi, Fanling Highway, planned road networks at San Tin Technopole, etc.) within the assessment area were included in the assessment using EPD Smart Air Modelling Platform v2.1 (SAMP v2.1) according to the latest Technical Notes on Air Quality Modelling on EPD's website and the EPD's User Guide for the ROAD Tool in Smart Air Modelling Platform. The modelled fleet is broken down into 18 vehicle classes based on the information in the Appendix 1 of Guideline on Modelling Vehicle Emissions published by EPD. The predicted 24-hour traffic flow and vehicle compositions at the identified roads was provided by the traffic consultant according to the methodology endorsed by Transport Department (TD) and are presented in **Appendix 3.2**.

3.6.2.3 EMFAC-HK v4.3 model was adopted in SAMP v2.1 to estimate the vehicular emission rates of NOx (i.e. initial NO + initial NO_2), RSP and FSP for vehicular emission arising from open road, and the adopted temperature and relative humidity data were extracted from the meteorological output in SAMP v2.1 of the PATH grids where the road segments are located. The monthly hour minimum temperature and relative humidity data was adopted for the prediction of short-term averages of NO_2, while the monthly hour average temperature and relative humidity data was adopted for the prediction of long-term averages of NO_2, and annual minimum temperature and relative humidity was adopted for prediction of short-term and long-term averages of RSP and FSP as conservative approach.

3.6.2.4 For calculations of start emissions for vehicle classes other than franchised buses, start emissions for each vehicle class were assumed to be directly proportional to Trips per Vehicle-kilometre-travelled ratio (Trip/VKT) with consideration of the proportion of local and rural roads within Hong Kong based on the Annual Traffic Census 2024 (ATC 2024) prepared by the TD. In the SAMP v2.1, default trip and VKT values from EMFAC-HK v4.3 model were used in the calculations and the proportion of local and rural roads within Hong Kong of 21.7% was calculated based on raw data from ATC 2024 provided by TD in accordance with the methodology described in the ATC 2024, with details provided in **Appendix 3.3**. The results calculated from raw data of ATC 2024 provided by TD are identical to ATC 2024. All rural and local roads within the assessment area are considered using the broad-brushed approach, including roads leading to the identified HGV parking sites. Zero Emission Vehicle (ZEV) Scenario has been considered in the open road emission calculation. The relevant parameters utilized in this assessment are provided in **Appendix 3.3.**

3.6.2.5 Direct noise remedies such as noise barriers are proposed under this Project. Secondary air quality impacts arising from the implementation of the proposed roadside noise mitigation measures, such as vertical noise barriers, and vertical noise barriers with canopies, were incorporated into the air quality model.

3.6.2.6 According to the *Technical Note for Modelling Vehicular Emission Using AERMOD (TN-AM01)*, with the installation of vertical noise barriers, all traffic pollutants generated from the mitigated road section are emitted from the top of the noise barriers. In the AERMOD model, the line source of mitigated road section would raise to the level of the vertical barrier. No correction or adjustment to the receiver heights was made in the model.

3.6.2.7 While for vertical noise barriers with cantilever canopies, it is assumed that dispersion of traffic pollutants would be in effect similar to physically shifting the mitigated road section towards the central divider. The traffic pollutants are assumed to emit from the top of the cantilever canopies. In the AERMOD model, the line source would horizontally shift to account for the covered extent, no adding the 3m mixing zone to the side of the road with the barrier. The line source of mitigated road section would also raise to the level of the barrier. No correction or adjustment to the receiver heights was made in the model.

Emission from the Existing Underpass near Koon Chun Sauce Factory and Proposed Tunnel

3.6.2.8 The portal emissions (initial NO, initial NO_2, RSP, and FSP) of the existing underpass near the existing Koon Chun Sauce Factory and the proposed tunnels are considered during the operation phase. The locations of portal emission are illustrated in **Figures 3.3**, **3.3.1**, **3.3.2**, and **3.3.3**.

3.6.2.9 For calculations of portal emissions from the existing underpass and proposed tunnels as well as noise enclosure, temperature and RH data was extracted from the SAMP of the concerned PATH grids. The annual minimum temperature and RH value (i.e. 6°C and 16%) was used in the calculation as a conservative approach. According to the Permanent International Association of Road Congress Report (PIARC, 2019), pollutants were assumed to eject from the portal as a portal jet such that 2/3 of the total emissions were dispersed within the first 50 m of the portal and 1/3 of the total emissions within the second 50 m. The detailed calculations of portal emissions were presented in **Appendix 3.4**.

Emissions associated with the existing and planned public transport interchange (PTI), bus and minibus termini, and heavy goods vehicles and coach parking sites

3.6.2.10 Vehicular emissions induced by existing and planned public transport interchange, bus and minibus termini, and heavy goods vehicles and coach parking sites were included in this assessment to avoid any underestimation of air quality impact. The vehicular emissions from these facilities within the 500m assessment area were calculated based on the start emission and running exhaust emission factors predicted by EMFAC-HK v4.3 model, cold idling emission factors from the *Calculation of Start Emissions in Air Quality Impact Assessment* published by EPD, warm idling emission factors from *Road Tunnels: Vehicle Emissions and Air Demand for Ventilation* published by World Road Association, and traffic data provided by the traffic consultant.

3.6.2.11 Start emissions from diesel vehicles fitted with selective catalytic reduction (SCR) devices were adjusted based on the idling emission and would be released over

total spread distance of 700 m from where the start takes place, while start emissions for LPG minibus, taxis and, private light bus were adjusted based on the idling emission and would be released over a total spread distance of 150 m from where the start takes place. Running exhaust and idling emissions would be released on the spot. For the identified PTI, termini, and parking sites, running exhaust and idling emissions from terminating and non-terminating vehicles, and adjusted start emission from terminating vehicles were considered for emissions inside the concerned facilities while the remaining adjusted start emission from terminating vehicles is considered for emissions outside the concerned facilities. The temperature and RH values for all concerned PATH grids were compared, and the minimum temperature and RH values (6°C and 16%) were adopted in the calculation as a conservative approach. The location of emission sources and detailed calculation of the emission are presented in **Figure 3.4** and **Appendix 3.5**, respectively.

3.6.2.12 According to the site survey conducted in May 2025, no operating mechanical ventilation was observed at the PTI at Park YOHO (ID02), Kwu Tung North Multi-Welfare Services Complex Terminus (ID18), carpark of Kim Logistics Co. Ltd. (ID10), warehouse / vehicle repair workshops along Pok Wai West Road (ID13), C-H Logistic (ID14), and Hichain Logistics Company Limited (ID15). Start emission inside the PTI / carpark would be modelled according to the existing layout of the PTI / carpark. As for the planned PTI at the private residential development under the planning application nos. Y/YL-NSW/8 and Y/YL-NSW/9 (ID19), mechanical ventilation would be deployed with the exhaust located at the eastern façade facing the San Tin Highway, according to the environmental assessment report of the planning application no. A/YL-NSW/9. Therefore, start emission inside this planned PTI would be modelled at the designated façade. Since there is no detailed design of the planned PTIs at Ngau Tam Mei (ID01) and Planned Transport Interchange Hub at San Tin / Lok Ma Chau Development Node (ID22), the modelling approach for start emission inside the PTI was referenced to the respective approved EIA Report (Development of Ngau Tam Mei Area (AEIAR-271/2025); San Tin / Lok Ma Chau Development Node (AEIAR-261/2024)). There is also no detailed design for the planned decked PTI at Public Housing Development at Sha Po (ID20) , planned decked PTI at Kwu Tung North (ID21), and planned PTI at A/YL-KTN/604 (ID24). Therefore, start emission inside the planned PTI was allocated at the egress of planned PTI as a conservative approach.

Determination of the Assessment Year

3.6.2.13 With reference to Clause 2.2 (d) of Appendix B of the EIA Study Brief (ESB-373/2025), the year with highest emission strength from road vehicles in the assessment area within the next 15 years upon commissioning of the Project would be selected as the assessment year.

3.6.2.14 The assessment years for open road vehicular emission was determined based on the highest emission strength from the road vehicles in the assessment area within the next 15 years upon commissioning of the Project. The Project will tentatively commence operation in Year 2036. Therefore, Years 2036 to 2046 and ultimate scenario were selected for determination of the assessment year. The vehicular burden of these years would be calculated and the results presented in **Table 3.11**. The worst assessment year is determined by the highest vehicular emission burden among NOx, RSP and FSP. According to the result in **Table 3.11**, the assessment year for NOx, RSP, and FSP is Year 2036.

Table 3.11 Emission Burden

Year	Emission Burden (tonne per year)		
	NOx	RSP	FSP
2036	**75.44**	**2.07**	**2.25**
2037	66.8	1.95	2.12
2038	62.1	1.87	2.04
2039	63.94	2.06	2.24
2040	59.92	1.98	2.15
2041	58.94	1.98	2.15
2042	55.48	1.92	2.09
2043	51.55	1.86	2.02
2044	45.32	1.65	1.8
2045	38.67	1.43	1.56
2046	39.22	1.45	1.58
Ultimate scenario[2]	41.91	1.62	1.75

Note:
[1] The highest emission burden of each pollutant is bold and underlined.
[2] Since the projected population in 2046 for the Northern Metropolis (NM) in 2021-based Territorial Population and Employment Data Matrix (TPEDM) is below NM's target of 2.5 million, and that the 2021-based TPEDM does not include design year beyond 2046, it is necessary to compile a planning dataset for the scenario (post-2046) that reflects the ultimate target population for the NM in order to ensure no under-estimation of traffic and vehicular burden for assessment. To develop the dataset for this ultimate scenario, it is proposed to adjust the population assumptions of relevant New Development Areas' populations to match their respective ultimate targets and align with NM's target population of 2.5 million. Given that this scenario represents NM's ultimate target population figures, the assessment year with maximum traffic projections within 15 years upon operation of the Project would be based on this ultimate scenario.

Industrial Emissions within 500m of the ST Section Project Boundary and Major Emissions within 4 km of the ST Section Project Boundary

3.6.2.15 Two industrial sources are associated with SP License. The SP license (L-8-006(2)) associated with gas work at Ngau Tam Mei Animal Waste Treatment Plant (AWTP) is located within the 500m assessment area, and the SP License (L-15-035(3)) associated with cement works from K. Wah Asphalt Ltd at Man Kam To Road (AP1) and is classified as a major point source situated within 4km of the ST Section. The locations are also presented in **Figure 3.5**.

3.6.2.16 Other chimneys identified within the 500m assessment were verified by questionnaires to the chimney owners/operators, the site survey in May 2025 and August 2025, approved EIA reports, and planning applications. The chimney parameters for all chimneys such as stack height, stack temperature, stack exit velocity, stack diameter, location and emission inventory were referred to the responses from the chimney owners/operators and the respective SP License records. The detailed calculation of industrial emission is presented in **Appendix 3.6**.

Concurrent Projects

3.6.2.17 As discussed in **Section 3.5.3.10**, cumulative impact from the concurrent projects such as Ngau Tam Mei New Development Area, San Tin Technopole, and Kwu Tung North New Development Area, etc. were considered in the assessment.

Dispersion Modelling and Modelling Approach

3.6.2.18 With reference to Section 3.5.3, Appendix B of the EIA Study Brief, and the EPD's *Guidelines for Local-Scale Air Quality Assessment Using Models*, the AERMOD model was employed to predict the potential air quality impact.

3.6.2.19 All open road vehicular emissions and portal emissions were modelled as "LINE" and "VOLUME" sources. The hourly initial NO, NO_2, RSP, and FSP emission factors of each road segment in grams per second per square meter were generated using SAMP v2.1. **Appendix 3.3** shows the model parameters in detail.

3.6.2.20 The existing HGV and coach parking sites (ID nos. 10, 13, 14, 15, and 18) are in open areas and do not have any mechanical ventilation for exhaust according to on-site observation. Thus, the emissions within these parking sites are modelled as "AREAPOLY" sources. As regard the PTI at Park YOHO (ID no. 02), it is situated under the residential building without mechanical ventilation as observed during a site visit. All emissions, within this PTI, are therefore allocated in the egress of the PTI and modelled as "VOLUME" source at that location as a conservative approach. Since the detailed designs of the planned PTIs (ID nos. 01, 19, 20, 21, 22 and 24) are not available at the time of preparation of this study, it is assumed that for ID nos. 01 and 22 the model setups are referenced to the approved EIA reports for the Development of Ngau Tam Mei Area (AEIAR-271/2025) and the San Tin / Lok Ma Chau Development Node (AEIAR-261/2024), respectively. For ID no. 19, based on the environmental assessment report of the approved planning application Y/YL-NSW/9, the covered transport layby would adopt mechanical ventilation with the exhaust located on the eastern façade, and the model has been setup accordingly. For ID nos. 20, 21 and 24, while all these PTIs are decked according to the respective gazette plan and PER report, no information on the ventilation method and exhaust location is available. Therefore, all emissions generated within these planned PTIs will be allocated in the egress of the PTI and modelled as "VOLUME" source at that location as a conservative approach. **Appendix 3.5** shows the model parameters in detail.

3.6.2.21 Hourly meteorological conditions including wind data, temperature, relative humidity, pressure, cloud cover and mixing height are extracted from the WRF meteorological data adopted in the PATH v3.0 model, obtained through SAMP v2.1. The meteorological data was inputted as on-site data into AERMET (the meteorological pre-processor of AERMOD). Surface characteristic parameters such as albedo, Bowen ratio and surface roughness are required in the AERMET (the meteorological pre-processor of AERMOD) and could be found in **Appendix 3.7**. The land use characteristics of the project area are customized according to the latest design of the related projects, while that of the surrounding are classified based on the Planning Department's 2024 land utilization map, and the AERMET output are generated by SAMP v2.1 for the AERMOD model run. Elevated terrain and urban mode in AERMOD were adopted for this assessment, with an urban population of 676,800 according to the latest 2024 population census.

3.6.3 Cumulative Impact

3.6.3.1 The PATH model output was added to the sum of the AERMOD model results sequentially on an hour-to-hour basis to derive the short-term and long-term cumulative impacts at the ASRs. Year 2035 PATH model output was adopted as the background concentrations as conservative approach and the assessment year is 2036. According to the result in **Table 3.11**, the assessment year for NOx, RSP, and FSP is Year 2036.

3.6.3.2 Time-averaged results, namely hourly, daily and annual, will be derived from the cumulative hour-by-hour results. For annual average, the sum of all hourly concentrations is divided by the number of hours during the year to obtain the annual-averaged concentration. The air quality impact upon ASRs will be evaluated by respective concentration limits specified in the AQO. For daily average, cumulative daily-averaged results at each ASR amongst 365 days are ranked by the concentration and the ranked daily-averaged concentrations are compared with the maximum allowable concentration to determine the number of exceedances throughout a year. The air quality impact upon ASRs was evaluated by number of exceedances per annum against the number of exceedances allowed as specified in the AQO or EIAO-TM for hourly and daily averaged result.

3.6.3.3 As suggested by "Guidelines on Assessing the 'TOTAL' Air Quality Impacts", an integrated modelling system, PATH v3.0 which is developed and maintained by EPD is applied to estimate the future background pollutant concentrations. The emission sources including those in Pearl River Delta Economic Zone, roads, marine, airport, power plants and industries within Hong Kong are all considered in the PATH v3.0 model. The year of the adopted PATH background data is 2035, which has the highest background concentration within the assessment periods.

Conversion of NO_x to NO_2

3.6.3.4 The conversion from NO_x to NO_2 is necessary to compare the operation phase cumulative results against AQOs. The conversion method follows the approaches stated in the 'Guidelines on Choice of Models and Model Parameters' published by EPD. Two approaches, namely Ozone Limiting Method and Jenkin Method, were adopted for the conversion from NO_x to NO_2.

Ozone Limiting Method for Short-term Cumulative NO_2 Assessment

3.6.3.5 For the short-term cumulative NO_2 assessment (i.e., predictions of hourly and daily average NO_2 concentrations), the Ozone Limiting Method (OLM) was adopted for conversion of NO from vehicle-related sources (i.e., emissions from open roads, portals, PTI, bus and minibus termini, and parking sites) and NOx from industrial source to NO_2 based on the predicted O_3 level from PATH v3.0. According to the Heathrow Airport EIA report, the initial NO_2/NOx ratios of industrial emission sources are 10%. The predicted initial NO concentrations from open roads, portals, PTI, bus and minibus termini, and heavy goods vehicles and coach parking sites, and 90% of the predicted initial NOx concentrations from industrial emissions were firstly added together on an hour-to-hour basis and OLM was applied subsequently to calculate NO_2 based on the predicted O_3 level from PATH v3.0. The NO_2/NOx conversion was calculated as follows:

$[NO_2]_{predicted} = [NO_2]_{vehicular} + 0.1 \times [NO_x]_{industrial} + MIN \{[NO]_{vehicular} + 0.9 \times [NOx]_{industrial}$, or $(46/48) \times [O_3]_{PATH}\}$
where

$[NO_2]_{predicted}$	is the predicted NO_2 concentration
$[NO_2]_{vehicular}$	is the sum of predicted initial NO_2 concentration from open roads, portals, PTI, bus and minibus termini, heavy goods vehicle and coach parking site
$[NO_x]_{industrial}$	is the sum of predicted initial NOx concentration from the industrial emission source
$[NO]_{vehicular}$	is the sum of predicted initial NO concentration from open roads, portals, PTI, bus and minibus termini, heavy goods vehicle and coach parking site
MIN	means the minimum of the two values within the brackets
$[O_3]_{PATH}$	is the representative O_3 from PATH concentration (from other contribution)
(46/48)	is the molecular weight of NO_2 divided by the molecular weight of O_3

Jenkin Method for Long-term Cumulative NO_2 Assessment

3.6.3.6 For the long-term cumulative NO_2 assessment (i.e., predictions of annual average NO_2 concentration). Jenkin method was adopted for the conversion of cumulative NOx to NO_2 by using the functional form of an annual mean of NO_2-to-NOx with reference to the *Review of Methods for NO to NO_2 Conversion in plumes at short ranges*[1]. The mentioned functional form is referenced from Jenkin, 2004a[2] and is presented as follows:

$$[NO2] = \frac{\left([\mathrm{NOx}] + [\mathrm{OX}] + \frac{\mathrm{J}}{\mathrm{k}}\right) - \sqrt{([NOx] + [OX] + \frac{J}{k})^2 - 4[NOx][OX]}}{2}$$

where

$[NO_2]$	is the NO_2 concentration
$[NO_x]$	is the NOx concentration
[OX]	is the sum of NO_2 concentration and O_3 concentration (i.e. $[OX] = [NO_2] + [O_3]$)
J	is the photolysis rate of NO_2
k	is the rate coefficient for reaction between NO and O_3

3.6.3.7 The above functional form was used to analyse the annual mean data obtained from nearest EPD's air quality monitoring stations including Yuen Long general station, North general station, Tap Mun general station, and three roadside stations (i.e., Causeway Bay, Central, and Mong Kok roadside station), in order to obtain the empirical relationship. The Yuen Long general station and North general station are chosen as the representative station due to their proximity to the Project. Tap Mun general station and the roadside stations were also included in order to cover a wider range of NOx concentration. The functional form curve would fit the annual mean data when [OX] = 86.94 μg/m^3 and J/k = 15.388 μg/m^3, as obtained from SAMP v2.1. The obtained functional form curve was adopted for the cumulative annual average NO_x to NO_2 conversion. As shown in **Appendix 3.8**, the curve is slightly higher than all the annual mean data obtained from AQMS and on-site

[1] *Environment Agency. 2007. Review of methods for NO to NO_2 conversion in plumes at short range. Prepared by Environmental Agency.*
[2] *Jenkin. 2004a. Analysis of sources and partitioning of oxidant in the UK – Part 1: The NOx-dependence of annual mean concentrations of nitrogen dioxide and ozone. Atmospheric Environment, 38, 5117-5129.*

measurement data, underestimation of the annual average NO_2 concentration is not expected.

3.7 Prediction and Evaluation of Environmental Impact

3.7.1 Construction Phase

3.7.1.1 As discussed in **Section 3.5.1.2**, construction of the ST Section involves site formation, site clearance, excavation, slope works, construction of administration buildings and satellite control buildings, blasting for tunnel construction, mucking out blasted rock and soil from drill-and-blast operations, material handling and backfilling, foundation works, handling and stockpiling of dusty materials. These works would involve earth moving activities and be affected by wind erosion and hence would cause potential air emissions.

3.7.1.2 Based on the latest design information, it is estimated that the total quantity of construction and demolition (C&D) materials which will be generated during construction phase of the Project would be approximately 5,669,590 m^3 (5,495,370 m^3 of inert materials and 174,220 m^3 of non-inert materials). The major construction activities that generate C&D materials would be carried out from Year 2027 to Year 2036. It should be noted that these quantities are initial estimates only and would need to be further reviewed and updated as the design progresses. Summary of the estimated generation of C&D materials (inert and non-inert) during the construction phase is presented in **Table 3.12**.

Table 3.12 Summary of Estimated Quantities of C&D Materials Generated during Construction Phase

Construction Activities	Estimated Quantities of Non-Inert C&D Materials (m^3) [1]	Estimated Quantities of Inert C&D Materials (m^3) [1]			
		Soft Material[2]	Broken Concrete	Rock[2]	Total
Site Clearance	174,220	0	94,160	0	94,160
Portal Site Formation	0	1,479,590	0	0	1,479,590
At-grade Road Site Formation	0	841,360	0	0	841,360
Tunnel Excavation	0	0	0	2,132,470	2,132,470
Pier Foundation Excavation	0	164,020	0	0	164,020
Road Surfacing	0	97,430	0	0	97,430
Building Site Formation	0	184,930	0	0	184,930
Underground Utilities Excavation	0	143,910	0	0	143,910
Wetland Compensation Excavation	0	357,500	0	0	357,500
Total	**174,220**	**3,268,740**	**94,160**	**2,132,470**	**5,495,370**

Note:
[1] The estimated quantities are rounded to the nearest 10 m^3.
[2] Bulking factor of 1.3 has been applied.

3.7.1.3 Air quality control measures stipulated in *Air Pollution Control (Construction Dust) Regulation* would be implemented to abate air emissions from the construction sites. Regular watering should be provided at the excavation and filling works, material

handling, and exposed areas. Stockpile areas should be covered with impervious sheets, as far as practicable. Haul roads (if any) should be paved and regularly wetted to suppress the fugitive dust emission caused by the travelling construction vehicles. Vehicles transporting dusty materials should be properly covered with mechanical covers or tarpaulin sheets to avoid any dust pickup by gust during travel. Wheel washing facility should also be provided at each exit of construction sites such that no residue on the body of construction vehicle would cause dust emission on public roads. It is understood that construction activities would not take place concurrently at the entire construction work area, but to be undertaken at multiple work fronts at different construction periods in order to minimize the potential air quality impact. With the implementation of appropriate mitigation measures and good site practices, the air emission from the construction works would be controlled.

3.7.1.4 Drill-and-blast method would be adopted for the construction of tunnels. For tunnel blasting, blast door would be closed, and ventilation system would be stopped during tunnel blasting to minimise potential dust emission from blasting activities. Dust filter of at least grade F8 in accordance with EN 779:2012 or grade E11 as per EN 1822 would be installed at the exhaust of the ventilation system, which is able to capture 90% to 95% of 0.4 μm particles. During open blasting activities, blast screen, blast cages, roof-over protective cover and blast mats will be implemented to minimise potential dust emission. The areas within 30m from the blasting area will be wetted with water prior to blasting and blasting shall not be carried out when the strong wind signal or tropical cyclone warning signal No. 3 or higher is hoisted, according to the *Air Pollution Control (Construction Dust) Regulation*. Where necessary, mist spraying measures will be implemented at the mucking out locations. Also, sequence of construction works would be adopted to avoid dusty construction activities to be carried out simultaneously close to ASRs as far as practicable. With all these proper designs, best site practices, and appropriate mitigation measures in place, adverse air quality impact due to blasting is not anticipated.

3.7.1.5 Fuel combustion from the use of powered mechanical equipment (PME) and vehicular emissions from construction vehicles during construction works are also sources of air emissions. According to *DEVB TC(W) No. 1/2015*, NRMMs with exempted label under the Air Pollution Control (NRMM) Regulation shall be avoided. In addition, the Contractor shall timely apply for the temporary electricity with a target that the necessary cables laying works can be completed before the commencement of construction works and avoid on-site use of diesel generator according to *DEVB TC (W) No. 13/2020*. The Contractor is not allowed to use exempted NRMM to ensure the adopted machines or non-road vehicle under the Project could meet the prescribed emission standards and requirements according to *DEVB TC(W) No. 1/2015*. The use of electrified NRMMs, which is unlikely to cause significant smoke and gaseous emissions, will also be adopted as far as practicable. Activities involving the use of non-electrical NRMMs should be located as far away as possible from the nearby ASRs, and careful scheduling of nearby construction works involving NRMMs could be managed by close coordination or collaboration among construction sites.

3.7.1.6 Based on the latest plant inventory list, the maximum number of non-road mobile machinery to be used at a time is 600. The Contractor shall follow the said regulations and technical circulars during the use of NRMMs. Hence, with the

implementation of the said Regulations and technical circulars, no adverse air quality impact is expected from the operation of the NRMMs.

3.7.1.7 Excavated materials would be transported by dump trucks out of the construction sites to the designated disposal outlets via the transportation routes stated in **Section 6**. A maximum of 1000 truck trips per day is required, and the tentative transport routings are (1) via San Tin Highway, Yuen Long Highway, New Territories Circular Road, Tuen Mun Road, San Tak Street, San Wu Lane, Tuen Mun Heung Sze Wui Road, Castle Peak Road – San Hui, Tuen Shing Street, Tuen Fat Road, Wong Chu Road, Lung Fu Road and Lung Mun Road to Tuen Mun Area 38 Fill Bank; or (2) via San Tin Highway, Fanling Highway, Heung Yuen Wai Highway Road, Wo Keng Shan Road to NENT Landfill; or (3) via San Tin Highway, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road to Chemical Waste Treatment Centre. The induced road traffic would cause vehicle emission along the transportation routes. The routings for trucks have been optimised to maximise the distance from nearby ASRs, the use of local roads have been avoided as far as practicable, and the truck traffic should avoid peak hours as far as practicable. Dump truck shall be equipped with water-tight container and mechanical cover, which would not cause fugitive dust emission on the open road. Environment-friendly diesel commercial vehicles (DCV) (i.e., Euro VI) should be adopted as much as practicable to reduce vehicular emissions arising from the operation of construction vehicles. With the implementation of these mitigation measures, it is anticipated that no adverse air quality impact would be caused by the transportation of spoils along the routes.

Consideration of Cumulative Impact

3.7.1.8 The construction works such as site clearance and site formation works associated with ‘Site Formation and Infrastructure Works for Public Housing Development at Sha Po, Yuen Long’ (Sha Po Project) by CEDD is approximately 40m away from the project boundary of the ST Section. As the construction period is under review, the contractors of both the ST Section and the Sha Po Project shall abide by the *Air Pollution Control (Construction Dust) Regulation* and liaise closely to avoid any heavy or dusty construction activities being carried out concurrently once the construction period has been finalized. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. Thus, adverse cumulative construction air quality impact is not expected.

3.7.1.9 In order to provide connection between the NMH and Ngau Tam Mei, the ST Section partially encroaches into the NTM Project by CEDD, which the tentative construction period Year 2027 to 2036 by phase. The construction of the ST Section would completely overlap with the construction of the NTM Project. Since the NTM Project occupies an area of 130 ha, it is expected that construction works such as site clearance, site formation, excavation, and backfilling, etc. will be carried out at multiple works areas. Hence, the Contractors of the ST Section and the NTM Project shall closely liaise to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. With the implementation of the air quality control measures stipulated in the *Air Pollution Control (Construction Dust) Regulation*, such as frequent watering at particularly dusty construction areas and areas close to ASRs, no adverse cumulative construction air quality impact is expected.

3.7.1.10 The ST Section also partially encroaches Northeast of the San Tin Technopole (STT) near Chau Tau, South of the STT near Siu Hom Tsuen, and Southeast of the STT by connecting to Road D5 of the STT at Pang Loon Tei. Being a large-scale new development area, the STT occupies 610 ha and the construction started in Year 2024 for the completion by Year 2039. According to the approved EIA report (AEIAR-261/2024), the site formation and site development works at the South and Southeast of the Project area (i.e., Siu Hom Tsuen and Pang Loon Tei respectively) of the main phase would tentatively commence in Year 2026 to Year 2038 and completely overlaps with the construction program of the ST Section, whilst the site formation and site development works at the Northeast of the Project Area of the initial phase would tentative commence in Year 2024 to 2031. The South of the STT is approximately 300m away from the ST Section and is separated by hilly terrain at Ngau Tam Shan. Therefore, cumulative construction air quality impact between the South of the STT and the ST Section is considered insignificant. Due to the large-scale project in nature, it is anticipated that the construction works would be carried out at multiple works areas. Hence, the Contractors of the ST Section and the STT Project shall closely liaise to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. With the implementation of the air quality control measures stipulated in the *Air Pollution Control (Construction Dust) Regulation*, such as frequent watering at particularly dusty construction areas and areas close to ASRs, no adverse cumulative construction air quality impact is anticipated.

3.7.1.11 The Kwu Tung North New Development Area (KTN NDA) partially encroaches the northern end of the ST Section. In order to provide connection to the KTN NDA, the ST Section connects the KTN NDA near Pak Shek Au. According to the approved EIA report on the 'North East New Territories New Development Areas' (AEIAR-175/2013), the construction works of KTN NDA has started in 2017 and is targeted to complete in Year 2031 tentatively. The assessment results predicted that, with the implementation of mitigation measures such as regular watering on exposed worksites and covering excavated or stockpile of dusty materials entirely by impervious sheeting, no adverse construction air quality impact is predicted. The Contractors of the ST Section and the KTN NDA shall closely liaise to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. With the implementation of the air quality control measures stipulated in the *Air Pollution Control (Construction Dust) Regulation*, such as frequent watering at particularly dusty construction areas and areas close to ASRs, no adverse cumulative construction air quality impact is anticipated.

3.7.1.12 The Northern Link (NOL) is a 10.7 km long railway which starts at the Kam Sheung Road Station and ends at the Kwu Tung Station with three intermediate stations at Au Tau, Ngau Tam Mei, and San Tin. Based on the approved EIA report (AEIAR-259/2024), the tentative construction period is 2025 to 2034, which overlaps with the construction period of the ST Section. Whilst most of the works of NOL is conducted underground, at-grade work including site formation, foundation works, excavation, etc. for the construction of Shui Mei Road Ancillary Building, Au Tau Station, Pok Wai Ancillary Building, Ngau Tam Mei Depot, Kwu Tung Road Ancillary Building, Pak Shek Au Ancillary building, and the Kwu Tung Station is located within the 500m assessment area. The at-grade works sites at Au Tau Station, Pok Wai

Ancillary Building, Ngau Tam Mei Depot, and Kwu Tung Road Ancillary Building encroaches the project boundary of the ST Section, whilst the at-grade works site at Shui Mei Road Ancillary Building is 250m away from the project boundary of the ST Section. Owing to the large construction footprint, it is anticipated that construction works would be carried out at multiple works areas. The Contractors of the ST Section and the NOL shall closely liaise to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. With the implementation of the air quality control measures stipulated in the *Air Pollution Control (Construction Dust) Regulation*, such as regular watering on construction works sites and exposed site surface, no adverse cumulative construction air quality impact is anticipated.

3.7.1.13 Part of the Ngau Tam Mei Water Treatment Works Extension (NTMWTWE) falls within the project boundary of the ST Section. Based on the latest information, the construction period of NTMWTWE is Year 2025 to 2030. Site clearance, construction of new filter building, civil/structure modification work associated with the chlorine building, laying of fresh water trunk mains, and excavation work for the construction of Chamber G would be carried out during the construction of the ST Section. The fresh water trunk main laying works would be constructed by sections with each section divided into sub-sections of approximately 30m long and 3m wide. Each sub-section with excavation works is expected to complete within 1 month. The modification works for Chamber G which would be underground works would be conducted by sections. With the implementation of the recommended mitigation measures such as water spraying during the mined excavation with drill and break construction works for modification works for Chamber G, procedures and requirements given in the *Air Pollution Control (Construction Dust) Regulation*, and good site practice as recommended in the approved EIA report (AEIAR-262/2024), no adverse construction air quality impact is anticipated. The contractors of the ST Section and NTMWTWE shall closely liaise to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. Therefore, no adverse cumulative construction air quality impact is predicted.

3.7.1.14 The Development of Wetland Conservation Parks System (WCPS) under the Northern Metropolis Development Strategy involves the development of five new wetland conservation parks (WCPs), namely Nam Sang Wai (NSW) WCP, Sam Po Shue (SPS) WCP and Hoo Hok Wai (HHW) WCP, Hong Kong Wetland Park Expansion Area (HKWP Expansion Area) and Sha Ling/Nam Hang Nature Park (SL/NH NP). Only the NSW WCP falls within the 500m assessment area and touches the project boundary of the ST Section. While the development plan of NSW WCP is under review, the contractor of the ST Section shall liaise with the contractor of NSW WCP once there is a confirmed development timeline and construction period to avoid any heavy or dusty construction activities being carried out at the same time. Therefore, no adverse cumulative construction air quality impact is anticipated.

3.7.1.15 The Tin Shui Wai (TSW) and Kwu Tung (KT) sections of the NMH are expected to commence construction during the tentative construction period of the ST Section. The TSW section partially encroaches the ST Section whilst the KT Section immediately connects to the ST Section. Being a 5.5 km-long and 5 km-long trunk

road respectively, the TSW and KT sections are anticipated to be constructed by sub-sections. While the tentative construction programmes of TSW, ST, and KT sections are under review, the contractors of the three sections of the NMH shall closely liaise to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. Therefore, no adverse cumulative construction air quality impact is predicted.

3.7.1.16 Several approved planning applications are located within the 500m assessment area which are listed in **Table 3.13**. The contractor of the ST Section closely liaises with the contractors of these developments to avoid any heavy or dusty construction activities being carried out concurrently. Should such work in proximity be unavoidable, the corresponding parties shall work closely to schedule the construction works at different periods of the day to minimize concurrent works. With the implementation of good site practices and mitigation measures stipulated in the *Air Pollution Control (Construction Dust) Regulation*, no adverse cumulative construction air quality impact is anticipated.

Table 3.13 Planning Applications within the 500m Assessment Area

Planning Application No.	Description	Tentative construction program	Approximate Separation Distance (m)
Land Sharing Pilot Scheme Application No. LSPS/002	Proposed Land Sharing Pilot Scheme for a Site at Various Lots in D.D. 105, Yuen Long, the New Territories	Completion in 2031	120
Y/YL-NSW/7	Proposed Comprehensive Residential Development to include Wetland Restoration Area at Wing Kei Tsuen, Nam Sang Wai, Yuen Long	Completion in 2030	<5
Y/YL-NSW/8	Proposed Amendments to the Nam Sang Wai Outline Zoning Plan from “OU(CDWRA)” Zone to OU(CDWRA)1” Zone at Various Lots in D.D.107 and Adjoining Government Land, West of Castle Peak Road - Tam Mi, Yuen Long	Completion in 2031	<5
Y/YL-NSW/9	Proposed Amendments to the Nam Sang Wai Outline Zoning Plan from “I(D)”, “OS”, and “Road” Zones to “R(E)” Zone at Lots 1910 RP (Part) and 1743 S.C RP (Part) in D.D.107 and Adjoining Government Land, West of Castle Peak Road - Tam Mi, Yuen Long	Completion in 2031	<5

Planning Application No.	Description	Tentative construction program	Approximate Separation Distance (m)
A/YL-NSW/274	Proposed Residential Development in "Undetermined" Zone in Lots 592 C1 A, 592 C4 and 1252 C IN D.D.115 (To be Known as Lot 1646 in D.D.115) at Tung Shing Lei, Yuen Long	No publicly available information	325
A/YL-NSW/314	Proposed Residential Development at Various Lots in D.D. 104, North of Kam Pok Road East, Pok Wai, Yuen Long	No publicly available information	240
A/YL-NSW/293	Proposed Residential Development at Various Lots in D.D. 103 and D.D. 105, Nam Sang Wai, Yuen Long	No publicly available information	20
A/YL-KTN/964	Proposed Residential Development at Lot 1071 in D.D.103, Ha Ko Po Tsuen, Kam Tin, Yuen Long	No publicly available information	110
A/YL-KTN/604 and A/YL-KTN/663	Proposed Residential Development at Various Lots in D.D. 107 and Adjoining Government Land, Kam Tin, Yuen Long	No publicly available information	440
A/YL-NSW/348 and A/YL-NSW/349	Proposed Social Welfare Facilities at Various Lots in D.D. 104, Nam Sang Wai	Completion in 2030	210
Y/NE-KTS/18	Proposed School at at Various Lots in D.D.94, D.D.98 and D.D100 and adjoining Government Land, in Kwu Tung South	Completion in 2036	15

3.7.1.17 With the implementation of careful scheduling of works, effective air quality control measures, good site practices, and close liaison with contractors of concurrent projects, no adverse cumulative air quality impact at the ASRs is anticipated during the construction of the Project and other concurrent projects. A comprehensive EM&A programme with RSP and FSP real-time monitoring should be conducted to ensure the proper implementation of measures and the compliance of AQOs during the construction of the Project.

3.7.2 Operation Phase

3.7.2.1 The cumulative air quality impact due to all identified emission sources within 500m assessment area and background concentration at representative ASRs in Year 2036 has been evaluated. The predicted cumulative NO_2, RSP, and FSP concentrations at the representative ASRs at all assessment heights in the worst-case Year of 2036 are summarised in **Table 3.14** and **Table 3.15**, respectively. The detailed assessment results of all relevant assessment heights of the representative ASRs are presented in **Appendix 3.9**.

3.7.2.2 The predictions showed that daily and annual averages of RSP and FSP, and hourly, daily and annual averages of NO_2 at representative ASRs would comply with the prevailing AQOs. No adverse cumulative air quality impact would be anticipated during the operation of the Project.

3.7.2.3 According to the discrete results, the worst affected level for ASRs would be in general 1.5 mAG for the majority of ASRs. The contour plots of RSP, FSP, and NO_2 at 1.5 mAG are provided in **Figures 3.6.1 to Figures 3.12.4**.

Table 3.14 Predicted Cumulative NO_2 Concentrations at Representative Air Sensitive Receivers in Year 2036

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m³)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		mAG	200	120	40
A001	Au Tau Youth Centre, 2C Castle Peak Road - Yuen Long	1.5 - 5	80.4 - 85.8	30.6 - 32.0	15.0 - 15.5
A002	HK Boy Scouts Association Yuen Long East District Centre	1.5 - 10	69.5 - 75.1	28.2 - 28.8	13.8 - 14.3
A003	Harmonic Villa Block A, 2 Yau Shin Street	1.5 - 10	66.3 - 104.1	28.4 - 43.3	15.3 - 20.8
A004	Shin Leung House, Long Shin Estate, 11 Yau Shin Street	1.5 - 60	61.4 - 65.1	25.4 - 26.9	12.5 - 13.5
A005	TWGHs C Y Ma Memorial College, 3 Yau Shin Street	1.5 - 10	66.9 - 88.9	28.5 - 34.2	14.4 - 16.9
A006	Temporary Structure in DD103 60 near Tung Shung Lei	1.5 - 5	71.2 - 71.3	28.9 - 29.0	14.6
A007	29 Tung Shing Lei	1.5 - 5	69.1 - 69.2	28.1	14.0 - 14.1
A008	32 Tung Shing Lei	1.5 - 5	69.7 - 69.8	28.3	14.2
A009	33 Tung Shing Lei	1.5 - 5	73.0 - 73.4	28.6	14.4
A010	37 Tung Shing Lei	1.5 - 10	72.8 - 76.1	28.7 - 29.3	14.4 - 14.7
A011	The Church of Jesus Christ of Latter-Day Saints, 1 Tung Shing Lei	1.5 - 10	75.9 - 80.2	30.0 - 31.9	15.2 - 16.0
A012	63 Tung Shing Lei	1.5 - 10	66.6 - 66.9	27.6 - 27.7	13.7 - 13.8
A013	Joyful House at DD103 1000	1.5 - 5	57.4	22.6 - 22.7	11.6
A014	Little Club House at DD103 346	1.5 - 5	57.4	22.9 - 23.0	11.7
A015	Anchors Academy Affiliated International Kindergarten cum Nursery	1.5 - 25	60.9 - 64.3	24.4 - 26.4	11.5 - 11.9
A016	Football Field at Anchors Academy Affiliated International Kindergarten cum Nursery	1.5	68.6	27.6	12.3
A017	AFCD Au Tau Fisheries Office	1.5 - 5	65.0 - 68.1	25.2 - 25.9	11.9 - 12.5
A018	Crescent Green Tower 6	1.5 - 50	59.2 - 62.4	22.6 - 25.3	11.0 - 11.7
A019	Crescent Green Tower 10	1.5 - 50	58.5 - 62.6	22.6 - 25.0	11.0 - 11.6
A020	3A Ha Ko Po Tsuen	1.5 - 10	60.8	25.2 - 25.4	11.7
A021	Village House in DD103 281 S.A ss.2 S.D in Ha Ko Po Tsuen	1.5 - 10	60.8	25.1 - 25.3	11.6 - 11.7
A022	Park Yoho Venezia Tower 7A	1.5 - 35	60.4 - 66.6	23.9 - 26.4	11.3 - 12.0
A023	Park Yoho Tower 8A, 18 Castle Peak Road - Tam Mi	1.5 - 45	59.1 - 61.0	22.9 - 24.2	11.0 - 11.4
A024	York International Kindergarten	1.5 - 10	61.4 - 63.2	24.9 - 25.5	11.5 - 11.8
A025	Riva Helorus Boulevard Tower 2	1.5 - 70	58.5 - 60.8	22.4 - 24.4	10.9 - 11.4
A026	Riva Helorus Boulevard Tower 3	1.5 - 70	58.5 - 60.8	22.4 - 24.7	10.9 - 11.5
A027	Riva Genova Boulevard Block 5	1.5 - 45	58.5 - 60.8	22.6 - 24.7	11.0 - 11.5

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		mAG	200	120	40
A028	Genova Boulevard Block 1, Riva, 1 Ying Ho Road	1.5 - 20	60.5	23.7 - 24.4	11.2 - 11.4
A029	Cervia Boulevard House 1, Riva, 1 Ying Ho Road	1.5 - 10	60.5 - 60.6	24.1 - 24.2	11.3
A030	Ferentino Boulevard Block 1, Riva, 1 Ying Ho Road	1.5 - 20	60.4 - 60.8	23.7 - 24.6	11.2 - 11.4
A031	Bevera Boulevard House 1, Riva, 1 Ying Ho Road	1.5 - 15	60.5	23.9 - 24.4	11.3
A032	Enna Boulevard Block 5, Riva, 1 Ying Ho Road	1.5 - 45	58.5 - 60.8	22.5 - 25.1	10.9 - 11.7
A033	Avio Boulevard House 1, Riva, 1 Ying Ho Road	1.5 - 15	60.4 - 60.5	24.1 - 24.4	11.3
A034	307 Fung Kat Heung	1.5 - 5	74.6 - 75.8	28.4 - 28.5	12.9 - 13.0
A035	Fung Kat Heung Basketball Court at GLA-YL 547	1.5	66.5	24.3	11.5
A036	Yue Cheong Yuen at DD107 569 near Fung Kat Heung Road	1.5 - 5	61.6	23.1	11.1
A039	15 Mo Fan Heung	1.5 - 5	78.9	27.9 - 28.0	12.7 - 12.8
A040	19 Mo Fan Heung	1.5 - 5	78.6 - 78.7	27.9	12.7 - 12.8
A041	79A Mo Fan Heung	1.5 - 5	71.2 - 71.4	25.2 - 25.5	11.9 - 12.1
A042	Park Yoho Sicilia Tower 1A	1.5 - 45	61.6 - 67.4	23.4 - 24.3	11.2 - 11.8
A043	Park Yoho Venezia Tower 2B	1.5 - 45	61.3 - 67.1	23.3 - 23.8	11.1 - 11.7
A044	Park Vista 1A Tower 11B	1.5 - 50	60.8 - 64.4	23.1 - 23.3	11.0 - 11.3
A045	Park Yoho Genova Tower 17	1.5 - 50	60.8 - 62.2	23.0 - 23.3	10.9 - 11.2
A046	Park Yoho Napoli Tower 28	1.5 - 40	60.8 - 62.3	22.9 - 23.0	11.0 - 11.1
A047	Park Yoho Milano Tower 31A	1.5 - 35	60.9 - 62.7	22.9 - 23.0	11.0 - 11.2
A048	7 Sha Po Tsuen	1.5 - 10	63.9 - 64.1	23.5 - 23.7	11.4 - 11.5
A049	Basketball Court at 400 Sha Po Tsuen	1.5	64.3	23.7	11.5
A050	Construction Sector Imported Labour Quarters S1	1.5 - 5	72.8	27.4 - 27.5	14.1 - 14.2
A051	Temporary Structure in DD104 4202 RP	1.5 - 5	65.2 - 65.5	23.7 - 23.8	11.7 - 11.8
A052	106 Mo Fan Heung	1.5 - 5	65.4 - 65.5	23.6	11.5
A053	171 Mo Fan Heung	1.5 - 5	64.5 - 64.7	23.2 - 23.3	11.4
A054	Merry Garden House 1	1.5 - 10	72.2 - 80.3	27.2 - 31.1	14.0 - 16.8
A055	166 Pok Wai	1.5 - 10	67.1 - 69.2	25.0 - 25.7	12.0 - 12.2
A056	169 Pok Wai	1.5 - 10	66.1 - 67.6	24.4 - 24.9	11.8 - 11.9
A057	170 Pok Wai	1.5 - 10	65.9 - 67.2	24.3 - 24.7	11.8 - 11.9
A058	172 Pok Wai	1.5 - 5	67.3 - 67.8	24.6 - 24.8	11.9 - 12.0
A059	177 Pok Wai	1.5 - 10	67.0 - 68.9	24.8 - 25.6	12.0 - 12.2

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		**mAG**	**200**	**120**	**40**
A060	181 Pok Wai	1.5 - 10	66.9 - 68.8	24.8 - 25.6	12.0 - 12.2
A061	230 Pok Wai	1.5 - 10	68.6 - 73.8	25.8 - 27.3	12.3 - 12.8
A062	119B Pok Wai (Grenville)	1.5 - 10	67.7 - 68.5	24.2 - 25.0	12.8 - 13.2
A063	138B-138 Pok Wai, Elite Garden	1.5 - 10	72.6 - 75.5	27.1 - 29.0	14.0 - 15.5
A064	138J Pok Wai, Elite Garden	1.5 - 15	68.4 - 79.0	25.8 - 30.9	12.9 - 16.7
A065	Village House in DD104 3722 S.B ss.6 S.D near Wing Kei Tsuen	1.5 - 10	68.0 - 68.1	24.5 - 25.1	12.8 - 13.0
A066	6E Wing Kei Tsuen	1.5 - 15	66.3 - 68.1	23.9 - 24.7	12.5 - 12.9
A067	Greenery Gardens 23, 3 Fairview Park Boulevard	1.5 - 10	73.8	26.2 - 26.4	13.8 - 13.9
A068	Meister House Tower 1, 1 Fairview Park Boulevard	1.5 - 10	77.0 - 78.1	27.3 - 29.1	14.7 - 16.0
A069	Village House in DD104 1895 S.B RP	1.5 - 5	68.7 - 69.1	26.4	13.2
A070	Village House in DD104 1831 near San Wai Tsuen	1.5 - 5	67.7 - 67.8	26.1	13.3
A071	Village House in DD104 1976 S.A near San Wai Tsuen	1.5 - 10	66.6 - 67.0	25.9	13.0
A072	221 San Wai Tsuen	1.5 - 10	69.3 - 71.8	26.3 - 27.0	13.2 - 13.5
A073	Wah On Villa Block 25, 138 San Wai Tsuen	1.5 - 10	66.4 - 66.8	25.8 - 25.9	13.0
A074	Ian Court Block 21	1.5 - 10	66.7 - 67.1	25.8 - 25.9	12.9 - 13.0
A075	Temporary Structure to the Southeast of Lamp Post VD3741 near Ching Yau Road	1.5 - 5	58.3 - 58.6	21.3 - 21.4	10.8
A076	Village House to the West of Lamp Post VD5437 near Ching Yau Road	1.5 - 5	57.0	21.3	11.0
A077	Village House in DD104 4700 near Ching Yau Road	1.5 - 5	58.9 - 128.3	20.3 - 33.3	10.1 - 16.9
A078	Village House to the North of Lamp Post V3154 near Ching Yau Road	1.5 - 10	52.6	19.4	9.3
A079	Village House to the Northwest of Lamp Post VD6536 near Ching Yau Road	1.5 - 5	52.3 - 52.4	19.5	9.4
A080	Village House to the Southwest of Lamp Post VD2284 near Ching Yau Road	1.5 - 5	59.3 - 59.4	21.6	10.8
A081	Dhamma Garden Meditation (Tawya) Centre at Ngau Tam Mei	1.5 - 5	59.1 - 59.2	21.6 - 21.7	10.9 - 11.0
A082	337 San Wai Tsuen	1.5 - 10	60.4 - 60.5	21.7	11.3
A083	63B San Wai Tsuen	1.5 - 10	58.5 - 59.8	21.1 - 21.4	10.7 - 10.9
A084	Wat Buddhahamaram at Ngau Tam Mei	1.5 - 5	58.7 - 59.1	21.5 - 21.7	10.8 - 10.9
A085	Village House	1.5 - 5	52.5	19.8	9.7
A086	House at Lot No.163 in D.D.104	1.5 - 5	52.7	19.5	9.4

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		**mAG**	**200**	**120**	**40**
A087	Yiu Shing Mo Temple	1.5 - 10	56.0	20.9 - 21.0	10.1
A088	Ngau Tam Mei Water Treatment Works Administration Building	1.5 - 10	47.9	16.8	8.2
A089	San Tin Barracks Block 77	1.5 - 5	50.4	19.5	9.0
A090	291 Chau Tau Tsuen	1.5 - 10	77.1 - 77.6	28.4 - 28.7	14.5 - 14.6
A091	Village House to the South of Lamp Post VD9164 in Cheung Lek	1.5 - 5	59.7 - 59.8	21.8	10.9 - 11.0
A092	Village House to the Southwest of Lamp Post VD6740 in Cheung Lek	1.5 - 5	53.9 - 54.1	20.2 - 20.3	9.7 - 9.8
A093	Village House to the Southwest of Lamp Post H3353 near Ki Lun Tsuen	1.5 - 10	62.6	21.9 - 22.4	11.1 - 11.3
A094	Village House to the Southeast of Lamp Post H3354 near Ki Lun Tsuen	1.5 - 10	62.1 - 62.2	21.9 - 22.1	11.1
A095	Village House to the South of Lamp Post VD0256 near Ki Lun Tsuen	1.5 - 10	61.8	21.8 - 21.9	11.1
A096	Village House to the South of Lamp Post VD0260 near Ki Lun Tsuen	1.5 - 10	61.2 - 61.4	21.7 - 21.8	11.0 - 11.1
A097	Village House to the Northwest of Lamp Post FA4049 near Ki Lun Tsuen	1.5 - 10	60.6 - 60.8	21.8 - 21.9	11.0 - 11.1
A098	Village type house to the north of Ki Lun Tsuen Playground	1.5 - 5	62.7 - 62.8	22.5 - 22.6	11.4
A099	Village type house to the north of Ki Lun Tsuen Playground	1.5 - 10	61.7 - 61.9	22.0 - 22.2	11.2
A100	Village type house to the south of Ki Lun Tsuen Playground	1.5 - 5	60.6 - 60.9	22.1	11.1
A101	Village type house to the south of Ki Lun Tsuen Playground	1.5 - 5	60.6 - 61.3	22.4 - 22.7	11.3 - 11.5
A102	Europa Garden Block B6	1.5 - 15	72.3 - 76.2	27.7 - 28.9	14.8 - 15.7
A103	Europa Garden Block B21	1.5 - 15	71.0 - 74.0	27.1 - 28.1	14.6 - 15.2
A104	Europa Garden Block B36	1.5 - 15	70.9 - 84.9	27.7 - 34.4	14.7 - 18.1
A105	Europa Garden Block A1	1.5 - 15	70.8 - 77.0	27.4 - 30.4	14.5 - 16.4
A106	Europa Garden Block A15	1.5 - 15	70.6 - 74.6	26.9 - 29.1	14.3 - 15.7
A107	Europa Garden Block A27	1.5 - 10	72.7 - 74.3	27.9 - 29.1	14.8 - 15.7
A108	Kwu Tung Market Shopping Centre	1.5 - 15	72.2 - 77.2	28.2 - 29.9	14.6 - 15.5
A109	Lady Ho Tung Welfare Centre Eco-Learn Institute	1.5 - 15	72.1 - 76.2	28.3 - 29.7	14.6 - 15.3
A110	10 Kwu Tung South Road	1.5 - 5	72.2 - 73.1	27.0 - 28.0	14.2 - 14.6
A111	31 Rigi Avenue, Valais	1.5 - 10	75.4 - 84.1	29.5 - 32.7	15.4 - 16.7
A112	9 Neuchatel Avenue, Valais	1.5 - 10	70.6 - 70.8	26.0 - 26.6	13.8 - 14.0
A113	21 Neuchatel Avenue, Valais	1.5 - 10	70.3 - 70.6	25.9 - 26.3	13.7 - 13.8
A114	21 Locarno Avenue, Valais	1.5 - 10	70.2 - 70.3	25.6 - 25.7	13.7

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		mAG	200	120	40
A115	12 Locarno Avenue, Valais	1.5 - 10	70.0 - 70.2	25.4 - 25.6	13.6 - 13.7
A116	Village type house at Ma Tso Lung	1.5 - 10	70.7 - 70.8	24.3 - 24.4	12.9 - 13.0
A117	Village type house at Tit Hang	1.5 - 5	71.8 - 72.3	24.5	13.2
A118	Ngai's Temple, 19 Cheung Lek	1.5 - 5	52.6 - 52.9	19.7	10.0
A119	15A Cheung Lek	1.5 - 10	52.4 - 53.0	19.7	10.0
A120	561 Hang Tau	1.5 - 10	60.5 - 61.0	21.8 - 22.0	11.5
A121	Grand Garden Block 5, 6a Hang Tau	1.5 - 15	59.0 - 59.4	21.8 - 21.9	11.4
A122	412 Hang Tau	1.5 - 10	58.8 - 59.2	21.8 - 21.9	11.4
A123	26 Gallen Boulevard, Valais	1.5 - 10	75.4 - 75.6	27.5 - 28.1	14.6 - 15.2
A124	9 Moritz Avenue, Valais	1.5 - 10	74.3 - 75.0	26.8 - 27.0	14.3 - 14.4
A125	17 Geneva East Avenue, Valais	1.5 - 10	72.3 - 72.7	26.1	14.0
A126	27 Geneva South Avenue, Valais	1.5 - 10	72.3 - 72.6	26.1	14.0
A127	Shum Residence	1.5 - 10	68.6 - 70.5	25.4 - 25.8	12.1 - 12.3
A128	Chan Ming Yam Tong	1.5 - 5	63.7 - 64.1	23.3	11.6
A129	Pok Oi Hospital	1.5 - 60	61.4 - 64.1	25.5 - 26.2	12.6 - 13.2
A130	Village Houses in D.D. 104	1.5 - 5	51.1	20.1	9.4 - 9.5
A131	Village House in D.D. 100	1.5 - 5	53.9 - 54.4	20.1 - 20.2	9.9
A132	Kwu Tung North Multi-welfare Services Complex	1.5 - 30	69.6 - 77.1	25.8 - 27.9	14.1 - 15.3
A133	Village House at Kam Tin	1.5 - 10	63.2 - 65.0	23.9 - 25.1	12.1 - 12.4
A134	Temporary Structure (Shop) at Nam Sang Wai	1.5	85.1	30.2	16.1
A135	Temporary Structure (Office) near Fung Kat Heung	1.5 - 5	85.7 - 91.6	33.1 - 35.7	14.6 - 15.7
A136	Village house at Mo Fan Heung	1.5 - 5	72.4 - 73.3	26.9 - 27.4	12.6 - 12.7
A137	Camp Site of Forest Pathway	1.5	63.3	23.1	11.4
A138	Sun Hing Farm	1.5	52.6	19.6	9.6
A139	Camp Site of Aqualand Camping	1.5	52.5	19.9	9.5
A140	Lohas Organic Farm	1.5	52.6	19.8	10.1
PA001	Proposed Public Housing Development at Sha Po	1.5 - 185	60.1 - 83.9	21.9 - 33.8	10.7 - 16.5
PA002	Proposed Government, Institution or Community Facility	1.5 - 185	60.1 - 78.1	21.9 - 30.2	10.7 - 14.3
PA003	Proposed Public Housing Development at Sha Po	1.5 - 185	60.1 - 71.8	21.9 - 25.5	10.7 - 12.1
PA004	Proposed Government, Institution or Community Facility	1.5 - 40	57.4 - 60.4	21.0 - 21.7	10.6 - 11.1

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		**mAG**	**200**	**120**	**40**
PA005	Proposed Private Residential atop Ngau Tam Mei Depot	1.5 - 210	56.7 - 61.0	20.6 - 21.9	10.2 - 11.3
PA006	Proposed Fire Station cum Ambulance Depot and Staff Quarters	1.5 - 120	56.7 - 61.6	20.8 - 22.6	10.4 - 11.5
PA007	Proposed Private Residential	1.5 - 165	56.7 - 62.9	20.8 - 23.5	10.4 - 12.2
PA008	Proposed Private Residential	1.5 - 170	56.7 - 63.2	20.8 - 22.6	10.4 - 12.2
PA009	Proposed Hospital	1.5 - 125	56.7 - 61.4	20.8 - 22.7	10.4 - 11.4
PA010	Proposed Hospital	1.5 - 125	56.7 - 60.4	21.0 - 21.7	10.4 - 11.1
PA011	Proposed Government, Institution or Community Facility	1.5 - 40	57.5 - 57.6	21.0 - 21.1	10.5
PA012	Proposed University	1.5 - 130	51.4 - 52.6	19.1 - 19.5	9.4 - 9.7
PA013	Proposed University	1.5 - 125	51.1 - 52.2	19.2 - 19.5	9.2 - 9.5
PA014	Proposed University	1.5 - 125	51.5 - 52.0	19.2 - 19.5	9.2 - 9.5
PA015	Proposed University	1.5 - 125	51.6 - 52.2	19.2 - 19.5	9.2 - 9.5
PA016	Proposed University	1.5 - 120	55.7 - 56.2	20.2 - 20.8	9.7 - 10.0
PA017	Planned Information and Technology - Zone 4	1.5 - 120	57.8 - 62.0	21.3 - 23.2	10.5 - 11.4
PA018	Planned Logistics, Storage and Warehouse	1.5 - 150	57.6 - 61.9	20.3 - 23.4	10.3 - 11.7
PA019	Planned Information and Technology - Zone 4	1.5 - 130	57.6 - 62.2	21.0 - 22.8	10.5 - 11.4
PA020	Planned Logistics, Storage and Warehouse	1.5 - 150	57.5 - 63.1	20.4 - 23.5	10.3 - 11.6
PA021	Planned Information and Technology - Zone 4	1.5 - 140	57.5 - 62.6	20.6 - 23.0	10.4 - 11.4
PA022	Planned Information and Technology - Zone 4	1.5 - 90	57.6 - 61.3	21.6 - 22.7	10.7 - 11.2
PA023	Planned Information and Technology - Zone 4	1.5 - 140	57.5 - 62.7	20.8 - 23.1	10.4 - 11.4
PA024	Planned Information and Technology - Zone 4	1.5 - 130	57.5 - 62.0	21.1 - 23.0	10.5 - 11.5
PA025	Planned Logistics, Storage and Warehouse	1.5 - 130	64.2 - 95.6	22.9 - 42.3	12.3 - 19.3
PA026	Planned Information and Technology - Zone 3	1.5 - 100	65.0 - 75.4	23.4 - 29.9	12.8 - 16.3
PA027	Planned Information and Technology - Zone 3	1.5 - 100	64.9 - 75.1	23.3 - 29.0	12.8 - 15.9
PA028	Planned Information and Technology - Zone 3	1.5 - 90	76.0 - 81.4	26.6 - 29.2	13.0 - 15.2
PA029	Planned Information and Technology - Zone 3	20 - 70	76.3 - 80.4	27.1 - 29.2	13.4 - 15.0
PA030	Planned Information and Technology - Zone 3	20 - 70	75.3 - 77.2	27.0 - 29.3	13.4 - 14.5
PA031	Planned Information and Technology - Zone 4	1.5 - 120	57.8 - 60.5	20.1 - 21.9	10.3 - 11.2
PA032	Proposed Government, Institution or Community Facility	1.5 - 120	67.4 - 76.8	23.8 - 30.3	12.7 - 16.9
PA033	Proposed Government, Institution or Community Facility	1.5 - 120	67.4 - 73.6	23.7 - 29.1	12.7 - 15.9
PA034	Proposed Secondary School	1.5 - 25	70.3 - 73.0	26.5 - 27.6	14.3 - 15.2

ASR	Description	Assessment Height	Predicted Pollutant Concentration (µg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		**mAG**	**200**	**120**	**40**
PA035	Proposed Hospital, Polyclinic and General Clinic / Health Centre	1.5 - 115	65.8 - 164.2	23.3 - 76.2	12.8 - 37.2
PA036	Proposed Hospital, Polyclinic and General Clinic / Health Centre	1.5 - 115	69.6 - 91.5	24.3 - 30.8	12.2 - 15.5
PA037	Proposed Government, Institution or Community Facility	1.5 - 115	69.9 - 82.2	24.1 - 29.6	12.2 - 15.2
PA038	Proposed Residential Building	1.5 - 100	70.7 - 75.4	25.5 - 27.5	12.7 - 14.1
PA039	Proposed Residential Building	1.5 - 100	70.7 - 75.2	25.3 - 27.4	12.7 - 14.0
PA040	Proposed Residential Building	1.5 - 120	70.3 - 73.7	24.2 - 26.9	12.2 - 13.9
PA041	Proposed Residential Building	1.5 - 120	70.5 - 73.1	24.0 - 26.7	12.2 - 13.9
PA042	Proposed Residential Building	1.5 - 110	70.4 - 75.2	24.0 - 27.7	12.2 - 14.2
PA043	Proposed Primary School	1.5 - 25	72.1 - 74.4	24.7 - 26.4	13.5 - 14.3
PA044	Proposed Government, Institution or Community Facility	1.5 - 100	70.5 - 74.2	23.8 - 26.2	12.4 - 14.1
PA045	Proposed Government, Institution or Community Facility	1.5 - 100	70.0 - 73.7	23.7 - 26.3	12.4 - 13.9
PA046	Proposed Business and Technology Park	1.5 - 50	69.7 - 75.9	26.2 - 28.0	13.9 - 15.3
PA047	Proposed Residential Building	1.5 - 115	70.2 - 76.3	23.5 - 28.1	12.4 - 14.6
PA048	Proposed Residential Building	1.5 - 115	70.2 - 75.6	23.4 - 27.4	12.4 - 14.4
PA049	Proposed Residential Building of Land Sharing Pilot Scheme Application No. LSPS/002	1.5 - 95	62.5 - 71.1	24.5 - 26.5	11.7 - 13.1
PA050	Proposed Residential Building of Land Sharing Pilot Scheme Application No. LSPS/002	1.5 - 95	62.5 - 71.8	24.5 - 26.7	11.7 - 13.3
PA051	Proposed Residential Building	1.5 - 105	70.4 - 75.4	24.1 - 28.0	12.8 - 14.4
PA052	Planned Information and Technology - Zone 3	1.5 - 70	75.2 - 76.5	27.1 - 29.5	13.4 - 14.4
PA053	Planned Residential Building in Planning Application no. Y/YL-NSW/7	1.5 - 50	60.5 - 68.5	22.3 - 25.1	11.3 - 13.1
PA054	Planned Residential Building in Planning Application no. Y/YL-NSW/7	1.5 - 50	60.6 - 67.8	22.4 - 24.5	11.3 - 12.8
PA055	Propose Residential Building in Planning Application no. Y/YL-NSW/8	1.5 - 110	61.3 - 98.3	22.1 - 39.4	11.0 - 20.6
PA056	Propose Residential Building in Planning Application no. Y/YL-NSW/8	1.5 - 110	60.6 - 101.5	22.1 - 36.3	10.9 - 18.9
PA057	Planned Residential Building in Planning Application no. Y/YL-NSW/9	1.5 - 115	60.4 - 94.9	22.1 - 33.7	10.9 - 17.6
PA058	Planned Residential Building in Planning Application no. Y/YL-NSW/9	1.5 - 110	60.4 - 99.6	22.2 - 34.3	10.9 - 17.8
PA059	Planned residential development under the Planning Application no. A/YL-NSW/274	1.5 - 70	61.4 - 70.6	25.5 - 28.2	12.5 - 14.4
PA060	Planned commercial, kindergarten, welfare facilities under the Planning Application no. A/YL-NSW/274	1.5 - 10	67.0 - 70.2	27.3 - 27.8	13.9 - 14.3

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)		
			19th Highest Hourly Average NO_2	10th Highest Daily Average NO_2	Annual Average NO_2
AQO		**mAG**	**200**	**120**	**40**
PA061	Planned residential development under the Planning Application no. A/YL-NSW/274	1.5 - 70	61.4 - 69.2	25.6 - 27.7	12.5 - 13.6
PA062	Planned residential development under the Planning Application no. A/YL-NSW/274	1.5 - 70	61.4 - 65.4	25.5 - 26.6	12.5 - 13.2
PA063	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	61.3 - 68.8	25.3 - 28.6	12.4 - 14.8
PA064	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	61.3 - 72.0	25.4 - 30.1	12.4 - 15.4
PA065	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 15	67.7 - 76.0	28.6 - 31.4	14.6 - 15.8
PA066	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	61.3 - 66.5	25.4 - 27.5	12.4 - 13.8
PA067	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	61.3 - 70.2	25.4 - 28.3	12.4 - 14.1
PA068	Planned residential development under the Planning Application no. A/YL-NSW/314	1.5 - 15	64.8 - 65.6	23.3 - 24.1	12.1 - 12.7
PA069	Planned residential development under the Planning Application no. A/YL-NSW/314	1.5 - 15	64.1	23.0 - 24.0	11.8 - 12.2
PA070	Planned residential development under the Planning Application no. A/YL-NSW/314	1.5 - 15	69.9 - 70.0	24.8 - 25.0	13.0 - 13.1
PA071	Planned School in Planning Application no. Y/YL-NSW/9	1.5 - 20	80.8 - 109.7	30.8 - 43.7	15.1 - 21.7
PA072	Planned Factory in Planning Application no. Y/YL-NSW/9	1.5 - 10	98.2 - 117.6	36.8 - 46.8	19.3 - 23.3
PA073	Planned NMH Administration Building	1.5 - 30	56.9 - 59.3	21.0 - 21.7	10.5 - 10.9
PA074	Planned residential development under the Planning Application no. A/YL-KTN/964	1.5 - 60	58.5 - 73.0	22.3 - 30.6	10.9 - 14.3

Table 3.15 Predicted Cumulative RSP and FSP Concentrations at Representative Air Sensitive Receivers in Year 2036

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
A001	Au Tau Youth Centre, 2C Castle Peak Road - Yuen Long	1.5 - 5	52.4 - 52.5	20.2	30.8	12.5 - 12.6
A002	HK Boy Scouts Association Yuen Long East District Centre	1.5 - 10	52.3	20.1 - 20.2	30.7	12.5
A003	Harmonic Villa Block A, 2 Yau Shin Street	1.5 - 10	52.3 - 52.8	20.2 - 20.5	30.7 - 31.0	12.6 - 12.8
A004	Shin Leung House, Long Shin Estate, 11 Yau Shin Street	1.5 - 60	52.2 - 52.3	20.1	30.6 - 30.7	12.4 - 12.5
A005	TWGHs C Y Ma Memorial College, 3 Yau Shin Street	1.5 - 10	52.3 - 52.6	20.2 - 20.3	30.7 - 30.8	12.5 - 12.6
A006	Temporary Structure in DD103 60 near Tung Shung Lei	1.5 - 5	52.3	20.2	30.7	12.5
A007	29 Tung Shing Lei	1.5 - 5	52.2	20.2	30.7	12.5
A008	32 Tung Shing Lei	1.5 - 5	52.2	20.2	30.7	12.5
A009	33 Tung Shing Lei	1.5 - 5	52.2	20.2	30.7	12.5
A010	37 Tung Shing Lei	1.5 - 10	52.2	20.2	30.7	12.5
A011	The Church of Jesus Christ of Latter-Day Saints, 1 Tung Shing Lei	1.5 - 10	52.2 - 52.3	20.2 - 20.3	30.7	12.6
A012	63 Tung Shing Lei	1.5 - 10	52.2	20.1 - 20.2	30.7	12.5
A013	Joyful House at DD103 1000	1.5 - 5	52.2	20.1	31.2	12.4
A014	Little Club House at DD103 346	1.5 - 5	52.2	20.1	31.2	12.4
A015	Anchors Academy Affiliated International Kindergarten cum Nursery	1.5 - 25	51.8	19.9 - 20.0	30.6	12.3
A016	Football Field at Anchors Academy Affiliated International Kindergarten cum Nursery	1.5	51.9	20.0	30.6	12.4
A017	AFCD Au Tau Fisheries Office	1.5 - 5	51.8 - 51.9	20.0	30.6	12.3 - 12.4
A018	Crescent Green Tower 6	1.5 - 50	51.8	19.9 - 20.0	30.6	12.3
A019	Crescent Green Tower 10	1.5 - 50	51.8	19.9	30.6	12.3
A020	3A Ha Ko Po Tsuen	1.5 - 10	51.8	19.9 - 20.0	30.6	12.3
A021	Village House in DD103 281 S.A ss.2 S.D in Ha Ko Po Tsuen	1.5 - 10	51.8	19.9	30.6	12.3

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
A022	Park Yoho Venezia Tower 7A	1.5 - 35	51.8	19.9 - 20.0	30.6	12.3
A023	Park Yoho Tower 8A, 18 Castle Peak Road - Tam Mi	1.5 - 45	51.8	19.9	30.6	12.3
A024	York International Kindergarten	1.5 - 10	51.8	19.9 - 20.0	30.6	12.3
A025	Riva Helorus Boulevard Tower 2	1.5 - 70	51.8	19.9	30.6	12.3
A026	Riva Helorus Boulevard Tower 3	1.5 - 70	51.8	19.9	30.6	12.3
A027	Riva Genova Boulevard Block 5	1.5 - 45	51.8	19.9	30.6	12.3
A028	Genova Boulevard Block 1, Riva, 1 Ying Ho Road	1.5 - 20	51.8	19.9	30.6	12.3
A029	Cervia Boulevard House 1, Riva, 1 Ying Ho Road	1.5 - 10	51.8	19.9	30.6	12.3
A030	Ferentino Boulevard Block 1, Riva, 1 Ying Ho Road	1.5 - 20	51.8	19.9	30.6	12.3
A031	Bevera Boulevard House 1, Riva, 1 Ying Ho Road	1.5 - 15	51.8	19.9	30.6	12.3
A032	Enna Boulevard Block 5, Riva, 1 Ying Ho Road	1.5 - 45	51.8	19.9	30.6	12.3
A033	Avio Boulevard House 1, Riva, 1 Ying Ho Road	1.5 - 15	51.8	19.9	30.6	12.3
A034	307 Fung Kat Heung	1.5 - 5	54.0	20.6	31.9	12.9
A035	Fung Kat Heung Basketball Court at GLA-YL 547	1.5	54.0	20.6	31.8	12.8
A036	Yue Cheong Yuen at DD107 569 near Fung Kat Heung Road	1.5 - 5	54.0	20.5	31.8	12.8
A039	15 Mo Fan Heung	1.5 - 5	54.0	20.6	31.9	12.9
A040	19 Mo Fan Heung	1.5 - 5	54.0	20.6	31.9	12.9
A041	79A Mo Fan Heung	1.5 - 5	54.0	20.6	31.8	12.9
A042	Park Yoho Sicilia Tower 1A	1.5 - 45	54.0	20.5 - 20.6	31.8	12.8 - 12.9
A043	Park Yoho Venezia Tower 2B	1.5 - 45	54.0	20.5 - 20.6	31.8	12.8
A044	Park Vista 1A Tower 11B	1.5 - 50	54.0	20.5 - 20.6	31.8	12.8
A045	Park Yoho Genova Tower 17	1.5 - 50	54.0	20.5	31.8	12.8
A046	Park Yoho Napoli Tower 28	1.5 - 40	54.0	20.5	31.8	12.8
A047	Park Yoho Milano Tower 31A	1.5 - 35	54.0	20.5	31.8	12.8
A048	7 Sha Po Tsuen	1.5 - 10	54.0	20.6	31.8	12.8
A049	Basketball Court at 400 Sha Po Tsuen	1.5	54.0	20.6	31.8	12.8
A050	Construction Sector Imported Labour Quarters S1	1.5 - 5	50.5	19.7	29.0	12.1
A051	Temporary Structure in DD104 4202 RP	1.5 - 5	50.3	19.6	28.9	12.0
A052	106 Mo Fan Heung	1.5 - 5	50.3	19.6	28.9	12.0
A053	171 Mo Fan Heung	1.5 - 5	50.3	19.6	28.9	12.0

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		mAG	75	30	37.5	15
A054	Merry Garden House 1	1.5 - 10	50.4 - 50.7	19.7 - 19.9	29.0 - 29.2	12.1 - 12.3
A055	166 Pok Wai	1.5 - 10	50.3	19.6	28.9	12.0
A056	169 Pok Wai	1.5 - 10	50.3	19.6	28.9	12.0
A057	170 Pok Wai	1.5 - 10	50.3	19.6	28.9	12.0
A058	172 Pok Wai	1.5 - 5	50.3	19.6	28.9	12.0
A059	177 Pok Wai	1.5 - 10	50.3	19.6	28.9	12.0
A060	181 Pok Wai	1.5 - 10	50.3	19.6	28.9	12.0
A061	230 Pok Wai	1.5 - 10	50.3	19.6 - 19.7	28.9	12.0 - 12.1
A062	119B Pok Wai (Grenville)	1.5 - 10	50.4	19.6 - 19.7	28.9 - 29.0	12.1
A063	138B-138 Pok Wai, Elite Garden	1.5 - 10	50.5 - 50.6	19.7 - 19.8	29.0 - 29.1	12.1 - 12.2
A064	138J Pok Wai, Elite Garden	1.5 - 15	50.3 - 50.7	19.7 - 19.9	29.0 - 29.2	12.1 - 12.3
A065	Village House in DD104 3722 S.B ss.6 S.D near Wing Kei Tsuen	1.5 - 10	50.4	19.6	28.9	12.0 - 12.1
A066	6E Wing Kei Tsuen	1.5 - 15	50.4	19.6	28.9	12.0 - 12.1
A067	Greenery Gardens 23, 3 Fairview Park Boulevard	1.5 - 10	53.9	20.7	31.5 - 31.6	12.9
A068	Meister House Tower 1, 1 Fairview Park Boulevard	1.5 - 10	53.9 - 54.0	20.7 - 20.8	31.6 - 31.7	13.0
A069	Village House in DD104 1895 S.B RP	1.5 - 5	53.8	20.6	31.5	12.9
A070	Village House in DD104 1831 near San Wai Tsuen	1.5 - 5	53.8	20.6	31.5	12.9
A071	Village House in DD104 1976 S.A near San Wai Tsuen	1.5 - 10	53.8	20.6	31.5	12.9
A072	221 San Wai Tsuen	1.5 - 10	53.8	20.6	31.5	12.9
A073	Wah On Villa Block 25, 138 San Wai Tsuen	1.5 - 10	53.8	20.6	31.5	12.9
A074	Ian Court Block 21	1.5 - 10	53.8	20.6	31.5	12.9
A075	Temporary Structure to the Southeast of Lamp Post VD3741 near Ching Yau Road	1.5 - 5	53.9	20.6	32.2	12.9
A076	Village House to the West of Lamp Post VD5437 near Ching Yau Road	1.5 - 5	54.0	20.7	32.2	12.9
A077	Village House in DD104 4700 near Ching Yau Road	1.5 - 5	52.1 - 52.7	20.3 - 20.8	31.6 - 32.0	12.6 - 13.1
A078	Village House to the North of Lamp Post V3154 near Ching Yau Road	1.5 - 10	52.0	20.3	31.5	12.6
A079	Village House to the Northwest of Lamp Post VD6536 near Ching Yau Road	1.5 - 5	52.0	20.3	31.5	12.6

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		mAG	75	30	37.5	15
A080	Village House to the Southwest of Lamp Post VD2284 near Ching Yau Road	1.5 - 5	53.9	20.6	32.2	12.9
A081	Dhamma Garden Meditation (Tawya) Centre at Ngau Tam Mei	1.5 - 5	53.9	20.7	32.2	12.9
A082	337 San Wai Tsuen	1.5 - 10	53.9	20.7	32.2	12.9
A083	63B San Wai Tsuen	1.5 - 10	53.9	20.6	32.1 - 32.2	12.9
A084	Wat Buddhahamaram at Ngau Tam Mei	1.5 - 5	53.9	20.6	32.2	12.9
A085	Village House	1.5 - 5	52.1	20.3	31.6	12.6
A086	House at Lot No.163 in D.D.104	1.5 - 5	52.0	20.3	31.5	12.6
A087	Yiu Shing Mo Temple	1.5 - 10	50.6	20.0	30.6	12.3
A088	Ngau Tam Mei Water Treatment Works Administration Building	1.5 - 10	51.8	19.8	30.5	12.2
A089	San Tin Barracks Block 77	1.5 - 5	51.1	20.0	30.1	12.3
A090	291 Chau Tau Tsuen	1.5 - 10	53.9	20.8	31.4	12.9
A091	Village House to the South of Lamp Post VD9164 in Cheung Lek	1.5 - 5	55.0	20.9	31.8	13.0
A092	Village House to the Southwest of Lamp Post VD6740 in Cheung Lek	1.5 - 5	52.3	20.4	31.0	12.7
A093	Village House to the Southwest of Lamp Post H3353 near Ki Lun Tsuen	1.5 - 10	55.0	20.9	31.9	13.0
A094	Village House to the Southeast of Lamp Post H3354 near Ki Lun Tsuen	1.5 - 10	55.0	20.9	31.9	13.0
A095	Village House to the South of Lamp Post VD0256 near Ki Lun Tsuen	1.5 - 10	55.0	20.9	31.9	13.0
A096	Village House to the South of Lamp Post VD0260 near Ki Lun Tsuen	1.5 - 10	55.0	20.9	31.9	13.0
A097	Village House to the Northwest of Lamp Post FA4049 near Ki Lun Tsuen	1.5 - 10	55.0	20.9	31.9	13.0
A098	Village type house to the north of Ki Lun Tsuen Playground	1.5 - 5	55.0	20.9	31.9	13.1
A099	Village type house to the north of Ki Lun Tsuen Playground	1.5 - 10	55.0	20.9	31.9	13.0

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
A100	Village type house to the south of Ki Lun Tsuen Playground	1.5 - 5	55.0	20.9	31.9	13.0
A101	Village type house to the south of Ki Lun Tsuen Playground	1.5 - 5	55.0	20.9	31.9	13.0 - 13.1
A102	Europa Garden Block B6	1.5 - 15	54.5 - 54.6	21.3	32.4 - 32.5	13.4
A103	Europa Garden Block B21	1.5 - 15	54.5	21.3	32.4	13.3 - 13.4
A104	Europa Garden Block B36	1.5 - 15	54.5 - 54.6	21.3 - 21.5	32.4 - 32.6	13.4 - 13.5
A105	Europa Garden Block A1	1.5 - 15	54.5	21.2 - 21.3	32.4	13.3 - 13.4
A106	Europa Garden Block A15	1.5 - 15	54.5	21.2 - 21.3	32.4	13.3 - 13.4
A107	Europa Garden Block A27	1.5 - 10	54.5	21.3	32.4	13.3 - 13.4
A108	Kwu Tung Market Shopping Centre	1.5 - 15	54.5	21.3	32.4	13.3 - 13.4
A109	Lady Ho Tung Welfare Centre Eco-Learn Institute	1.5 - 15	54.4 - 54.5	21.3	32.4	13.3 - 13.4
A110	10 Kwu Tung South Road	1.5 - 5	54.4	21.2	32.3	13.3
A111	31 Rigi Avenue, Valais	1.5 - 10	54.5 - 54.6	21.3 - 21.4	32.4 - 32.5	13.4 - 13.5
A112	9 Neuchatel Avenue, Valais	1.5 - 10	54.4	21.2	32.3	13.3
A113	21 Neuchatel Avenue, Valais	1.5 - 10	54.4	21.2	32.3	13.3
A114	21 Locarno Avenue, Valais	1.5 - 10	54.4	21.2	32.3	13.3
A115	12 Locarno Avenue, Valais	1.5 - 10	54.4	21.2	32.3	13.3
A116	Village type house at Ma Tso Lung	1.5 - 10	54.3	21.1	32.2	13.2
A117	Village type house at Tit Hang	1.5 - 5	54.3	21.1	32.2	13.2
A118	Ngai's Temple, 19 Cheung Lek	1.5 - 5	54.5	20.7	32.7	13.0
A119	15A Cheung Lek	1.5 - 10	54.5	20.7	32.6	13.0
A120	561 Hang Tau	1.5 - 10	55.6	21.3	33.8	13.4
A121	Grand Garden Block 5, 6a Hang Tau	1.5 - 15	55.6	21.3	33.8	13.4
A122	412 Hang Tau	1.5 - 10	55.6	21.3	33.8	13.4
A123	26 Gallen Boulevard, Valais	1.5 - 10	54.8 - 54.9	21.3	33.0 - 33.1	13.4
A124	9 Moritz Avenue, Valais	1.5 - 10	54.8	21.3	33.0	13.4
A125	17 Geneva East Avenue, Valais	1.5 - 10	54.8	21.3	33.0	13.4
A126	27 Geneva South Avenue, Valais	1.5 - 10	54.8	21.3	33.0	13.4
A127	Shum Residence	1.5 - 10	54.0	20.6	31.8	12.9
A128	Chan Ming Yam Tong	1.5 - 5	50.3	19.6	28.9	12.0
A129	Pok Oi Hospital	1.5 - 60	52.2	20.1	30.6	12.5

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
A130	Village Houses in D.D. 104	1.5 - 5	51.1	20.0	30.2	12.3
A131	Village House in D.D. 100	1.5 - 5	52.3	20.4 - 20.5	31.0	12.7
A132	Kwu Tung North Multi-welfare Services Complex	1.5 - 30	54.5	21.2 - 21.3	32.4	13.3 - 13.4
A133	Village House at Kam Tin	1.5 - 10	51.9	20.0	30.6	12.3
A134	Temporary Structure (Shop) at Nam Sang Wai	1.5	54.5	20.8	32.1	13.1
A135	Temporary Structure (Office) near Fung Kat Heung	1.5 - 5	54.1	20.7	32.0	12.9 - 13.0
A136	Village house at Mo Fan Heung	1.5 - 5	50.3	19.6	28.9	12.0
A137	Camp Site of Forest Pathway	1.5	50.3	19.6	28.9	12.0
A138	Sun Hing Farm	1.5	52.0	20.3	31.6	12.6
A139	Camp Site of Aqualand Camping	1.5	52.3	20.4	31.0	12.7
A140	Lohas Organic Farm	1.5	54.5	20.7	32.7	13.0
PA001	Proposed Public Housing Development at Sha Po	1.5 - 185	54.0 - 54.1	20.5 - 20.7	31.8 - 32.0	12.8 - 13.0
PA002	Proposed Government, Institution or Community Facility	1.5 - 185	54.0 - 54.1	20.5 - 20.6	31.8 - 31.9	12.8 - 12.9
PA003	Proposed Public Housing Development at Sha Po	1.5 - 185	54.0	20.5 - 20.6	31.8	12.8 - 12.9
PA004	Proposed Government, Institution or Community Facility	1.5 - 40	53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA005	Proposed Private Residential atop Ngau Tam Mei Depot	1.5 - 210	53.8 - 53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA006	Proposed Fire Station cum Ambulance Depot and Staff Quarters	1.5 - 120	53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA007	Proposed Private Residential	1.5 - 165	53.8 - 53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA008	Proposed Private Residential	1.5 - 170	53.8 - 53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA009	Proposed Hospital	1.5 - 125	53.8 - 53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA010	Proposed Hospital	1.5 - 125	53.8 - 53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA011	Proposed Government, Institution or Community Facility	1.5 - 40	53.8 - 53.9	20.6	32.1	12.9
PA012	Proposed University	1.5 - 130	52.0 - 52.1	20.3	31.5 - 31.6	12.6
PA013	Proposed University	1.5 - 125	52.0	20.3	31.5 - 31.6	12.6
PA014	Proposed University	1.5 - 125	52.0	20.3	31.5	12.6
PA015	Proposed University	1.5 - 125	52.0	20.3	31.5	12.6
PA016	Proposed University	1.5 - 120	50.6	20.0	30.5 - 30.6	12.3

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
PA017	Planned Information and Technology - Zone 4	1.5 - 120	53.9	20.5 - 20.6	31.2 - 31.3	12.7 - 12.8
PA018	Planned Logistics, Storage and Warehouse	1.5 - 150	53.9	20.5 - 20.6	31.2 - 31.3	12.7 - 12.8
PA019	Planned Information and Technology - Zone 4	1.5 - 130	53.9	20.5 - 20.6	31.2 - 31.3	12.7 - 12.8
PA020	Planned Logistics, Storage and Warehouse	1.5 - 150	53.9	20.5 - 20.6	31.2 - 31.3	12.7 - 12.8
PA021	Planned Information and Technology - Zone 4	1.5 - 140	53.9	20.5 - 20.6	31.2 - 31.3	12.7 - 12.8
PA022	Planned Information and Technology - Zone 4	1.5 - 90	53.9	20.6	31.3	12.8
PA023	Planned Information and Technology - Zone 4	1.5 - 140	53.9	20.5 - 20.6	31.2 - 31.3	12.7 - 12.8
PA024	Planned Information and Technology - Zone 4	1.5 - 130	53.9	20.5 - 20.6	31.3	12.7 - 12.8
PA025	Planned Logistics, Storage and Warehouse	1.5 - 130	53.0 - 53.2	20.4 - 20.7	30.9 - 31.1	12.7 - 12.9
PA026	Planned Information and Technology - Zone 3	1.5 - 100	53.1 - 53.2	20.5 - 20.6	30.9 - 31.1	12.7 - 12.8
PA027	Planned Information and Technology - Zone 3	1.5 - 100	53.1 - 53.2	20.5 - 20.6	30.9 - 31.0	12.7 - 12.8
PA028	Planned Information and Technology - Zone 3	1.5 - 90	53.7 - 53.9	20.7 - 20.8	31.3 - 31.5	12.8 - 12.9
PA029	Planned Information and Technology - Zone 3	20 - 70	53.8	20.7 - 20.8	31.4 - 31.5	12.8 - 12.9
PA030	Planned Information and Technology - Zone 3	20 - 70	53.8 - 53.9	20.7 - 20.8	31.3 - 31.4	12.8 - 12.9
PA031	Planned Information and Technology - Zone 4	1.5 - 120	55.0	20.9	31.8 - 31.9	13.0
PA032	Proposed Government, Institution or Community Facility	1.5 - 120	54.4 - 54.6	21.2 - 21.4	32.3 - 32.5	13.3 - 13.5
PA033	Proposed Government, Institution or Community Facility	1.5 - 120	54.4 - 54.6	21.2 - 21.3	32.3 - 32.5	13.3 - 13.4
PA034	Proposed Secondary School	1.5 - 25	54.5	21.2 - 21.3	32.4	13.3 - 13.4
PA035	Proposed Hospital, Polyclinic and General Clinic / Health Centre	1.5 - 115	54.4 - 54.5	21.2 - 21.3	32.3 - 32.4	13.3 - 13.4
PA036	Proposed Hospital, Polyclinic and General Clinic / Health Centre	1.5 - 115	54.2	21.1 - 21.2	32.1 - 32.2	13.1 - 13.2
PA037	Proposed Government, Institution or Community Facility	1.5 - 115	54.2 - 54.3	21.1 - 21.2	32.1 - 32.2	13.1 - 13.2
PA038	Proposed Residential Building	1.5 - 100	54.2	21.1	32.1	13.1 - 13.2
PA039	Proposed Residential Building	1.5 - 100	54.2	21.1	32.1	13.1 - 13.2
PA040	Proposed Residential Building	1.5 - 120	54.2	21.1	32.1	13.1 - 13.2
PA041	Proposed Residential Building	1.5 - 120	54.2	21.1	32.1	13.1 - 13.2
PA042	Proposed Residential Building	1.5 - 110	54.2	21.1	32.1 - 32.2	13.1 - 13.2
PA043	Proposed Primary School	1.5 - 25	54.3	21.1	32.2	13.2

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
PA044	Proposed Government, Institution or Community Facility	1.5 - 100	54.2 - 54.3	21.1	32.1 - 32.2	13.1 - 13.2
PA045	Proposed Government, Institution or Community Facility	1.5 - 100	54.2 - 54.3	21.1	32.1 - 32.2	13.1 - 13.2
PA046	Proposed Business and Technology Park	1.5 - 50	54.8 - 55.0	21.3 - 21.4	33.0 - 33.1	13.4 - 13.5
PA047	Proposed Residential Building	1.5 - 115	54.6 - 54.7	21.2 - 21.3	32.4 - 32.5	13.2 - 13.3
PA048	Proposed Residential Building	1.5 - 115	54.6 - 54.7	21.2 - 21.3	32.4 - 32.5	13.2 - 13.3
PA049	Proposed Residential Building of Land Sharing Pilot Scheme Application No. LSPS/002	1.5 - 95	54.3 - 54.4	20.7	31.8 - 31.9	12.9 - 13.0
PA050	Proposed Residential Building of Land Sharing Pilot Scheme Application No. LSPS/002	1.5 - 95	54.3 - 54.4	20.7	31.8 - 31.9	12.9 - 13.0
PA051	Proposed Residential Building	1.5 - 105	54.6 - 54.7	21.2 - 21.3	32.4 - 32.5	13.3
PA052	Planned Information and Technology - Zone 3	1.5 - 70	53.8 - 53.9	20.7 - 20.8	31.3 - 31.4	12.8 - 12.9
PA053	Planned Residential Building in Planning Application no. Y/YL-NSW/7	1.5 - 50	50.3 - 50.4	19.6	28.9	12.0 - 12.1
PA054	Planned Residential Building in Planning Application no. Y/YL-NSW/7	1.5 - 50	50.3 - 50.4	19.6	28.9	12.0
PA055	Propose Residential Building in Planning Application no. Y/YL-NSW/8	1.5 - 110	50.3 - 50.9	19.5 - 20.0	28.9 - 29.3	12.0 - 12.4
PA056	Propose Residential Building in Planning Application no. Y/YL-NSW/8	1.5 - 110	54.0 - 54.5	20.5 - 20.9	31.8 - 32.2	12.8 - 13.2
PA057	Planned Residential Building in Planning Application no. Y/YL-NSW/9	1.5 - 115	54.0 - 54.3	20.5 - 20.8	31.8 - 32.1	12.8 - 13.1
PA058	Planned Residential Building in Planning Application no. Y/YL-NSW/9	1.5 - 110	54.0 - 54.3	20.5 - 20.9	31.8 - 32.1	12.8 - 13.1
PA059	Planned residential development under the Planning Application no. A/YL-NSW/274	1.5 - 70	52.1 - 52.2	20.1 - 20.2	30.6	12.5 - 12.6
PA060	Planned commercial, kindergarten, welfare facilities under the Planning Application no. A/YL-NSW/274	1.5 - 10	52.2	20.2	30.6	12.5 - 12.6
PA061	Planned residential development under the Planning Application no. A/YL-NSW/274	1.5 - 70	52.1 - 52.2	20.1 - 20.2	30.6	12.5

ASR	Description	Assessment Height	Predicted Pollutant Concentration (μg/m^3)			
			10th Highest Daily Average RSP	Annual Average RSP	19th Highest Daily Average FSP	Annual Average FSP
AQO		**mAG**	**75**	**30**	**37.5**	**15**
PA062	Planned residential development under the Planning Application no. A/YL-NSW/274	1.5 - 70	52.1 - 52.2	20.1	30.6	12.4 - 12.5
PA063	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	52.1 - 52.4	20.1 - 20.2	30.6 - 30.7	12.4 - 12.6
PA064	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	52.1 - 52.4	20.1 - 20.2	30.6 - 30.8	12.4 - 12.6
PA065	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 15	52.3 - 52.4	20.2	30.7 - 30.8	12.5 - 12.6
PA066	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	52.1 - 52.3	20.1 - 20.2	30.6 - 30.7	12.4 - 12.5
PA067	Planned residential development under the Planning Application no. A/YL-NSW/293	1.5 - 135	52.1 - 52.3	20.1 - 20.2	30.6 - 30.7	12.4 - 12.5
PA068	Planned residential development under the Planning Application no. A/YL-NSW/314	1.5 - 15	50.3 - 50.4	19.6	28.9	12.0
PA069	Planned residential development under the Planning Application no. A/YL-NSW/314	1.5 - 15	50.3	19.6	28.9	12.0
PA070	Planned residential development under the Planning Application no. A/YL-NSW/314	1.5 - 15	53.8	20.6	31.5	12.9
PA071	Planned School in Planning Application no. Y/YL-NSW/9	1.5 - 20	54.1 - 54.4	20.7 - 21.1	32.0 - 32.2	13.0 - 13.3
PA072	Planned Factory in Planning Application no. Y/YL-NSW/9	1.5 - 10	54.4 - 54.6	20.9 - 21.1	32.2 - 32.3	13.2 - 13.4
PA073	Planned NMH Administration Building	1.5 - 30	53.9	20.6 - 20.7	32.1 - 32.2	12.9
PA074	Planned residential development under the Planning Application no. A/YL-KTN/964	1.5 - 60	51.8	19.9 - 20.0	30.6	12.3 - 12.4

3.8 Mitigation Measures

3.8.1 Construction Phase

3.8.1.1 Air quality control measures stipulated in *Air Pollution Control (Construction Dust) Regulation* and good site practices listed below should be carried out to further minimise construction air quality impact:

- Regular watering to reduce dust emissions from exposed site surfaces and unpaved roads, particularly during dry weather;
- Frequent watering for particularly dusty construction areas and areas close to ASRs;
- For the works sites close to the ASRs with a separation distance less than 10 m as shown in **Appendix 3.1**, provide hoardings of not less than 3.5 m high from ground level along the site boundary; for the other works sites in general, provide hoarding of not less than 2.4 m high from ground level along site boundary except for site entrance or exit;
- Avoid positioning material stockpiling areas, major haul roads and dusty works within the construction site close to concerned ASRs;
- Avoid unnecessary exposed earth;
- Locate all the dusty activities away from any nearby ASRs as far as practicable;
- Open stockpiles shall be avoided or covered. Where possible, prevent placing dusty material storage piles near ASRs;
- Tarpaulin covering of all dusty vehicle loads transported to, from and between site locations;
- Establishment and use of vehicle wheel and body washing facilities at the exit points of the site;
- Where possible, routing of vehicles and positioning of construction plant should be at the maximum possible distance from ASRs;
- Imposition of speed controls for vehicles on site haul roads; and
- Instigation of an EM&A programme with continuous construction dust monitoring to monitor the construction process in order to enforce controls and modify method of work if dusty conditions arise.

3.8.1.2 Guidelines stipulated in EPD's *Recommended Pollution Control Clauses for Construction Contracts* should be incorporated in the contract document to abate construction air quality impacts. These clauses include:

- The Contractor shall observe and comply with APCO and its subsidiary regulation, particularly the *Air Pollution Control (Open Burning) Regulation*, *Air Pollution Control (Construction Dust) Regulation*, *Air Pollution Control (Non-road Mobile Machinery)(Emission) Regulation*, *Air Pollution Control (Fuel Restriction) Regulation*, and *Air Pollution Control (Smoke) Regulation*;

- In addition to the statutory requirements of the Regulations, the Contractor of the public works contracts shall also observe the requirements as set out in the government circulars, including *DEVB's TC No, 13/2020 (Timely Application of Temporary Electricity and Water Supply for Public Works Contracts and Wider Use of Electric Vehicles in Public Works Contracts)* and *DEVB's TC No. 1/2015 (Emissions Control of NRMM in Capital Works Contracts of Public Works)*;

- The Contractor shall undertake at all times to prevent dust nuisance as a result of the construction activities, and minimise the emission of air pollutants from construction plant and equipment;

- The Contractor shall ensure that there will be adequate water supply / storage for dust suppression;

- The Contractor shall devise and arrange methods of working and carrying out the works in such a manner so as to minimise dust impact on the surrounding environment, and shall provide experienced personnel with suitable training to ensure that these methods are implemented properly;

- For better smoke control, the Contractor shall not use diesel hammer for percussive piling; and

- Before the commencement of any work, the Engineer may require the methods of working, plant, equipment and air pollution control system to be used on the site to be made available for inspection and approval to ensure that they are suitable for the project.

3.8.1.3 In order to help reduce carbon emission and pollution, timely application of temporary electricity and water supply would be made and electric vehicles would be adopted in accordance with *DEVB TC(W) No. 13/2020 – Timely Application of Temporary Electricity and Water Supply for Public Works Contracts and Wider Use of Electric Vehicles in Public Works Contracts* in the Project.

3.8.1.4 To minimise the exhaust emission from NRMMs during the construction phase, *DEVB TC(W) No. 1/2015 - Emissions Control of NRMM in Capital Works Contracts of Public Works* shall be followed, and the following measures should be applied:

- Connect construction plant and equipment to main electricity supply and avoid use of diesel generators and diesel-powered equipment as far as practicable;

- Avoid exempted NRMMs; and

- Deploy electrified NRMMs as far as practicable

3.8.1.5 For the tunnelling works by drill-and-blast, a blast door should be installed at the opening to avoid the escape of fugitive dust from the blasting. The blast door should remain closed and ventilation with dust filter of at least grade F8 in accordance with EN 779:2012 or grade E11 as per EN 1822 would be installed at the exhaust of the ventilation system, which is able to capture 90% to 95% of 0.4 μm particles. Water spaying should be applied to facilitate dust settlement. The areas within 30m from the blasting area should be wetted with water prior to blasting and blasting shall not be carried out when the strong wind signal or tropical cyclone warning signal No. 3 or higher is hoisted.

3.8.2 Operation Phase

3.8.2.1 No adverse air quality impact is anticipated during the operation phase of the Project, thus mitigation measure is deemed not necessary.

3.9 Evaluation of Residual Impact

3.9.1 Construction Phase

3.9.1.1 With the implementation of the air quality control measures specified in *Air Pollution Control (Construction Dust) Regulation* together with the recommended good site practices such as regular watering on the works areas, exposed surface and paved road, no residual impact would be expected from the construction of the Project.

3.9.2 Operation Phase

3.9.2.1 No residual impact is expected during the operation phase of the Project.

3.10 Environmental Monitoring and Audit

3.10.1 Construction Phase

3.10.1.1 Environmental Monitoring and Audit (EM&A) for potential air quality impacts are recommended during the construction phase of the Project so as to check compliance with legislative requirements and ensure the air quality control measures are implemented in order. Details of the monitoring and audit programme are presented in a stand-alone EM&A Manual.

3.10.2 Operation Phase

3.10.2.1 No adverse air quality impact would be generated during the operation phase of the Project. No EM&A would be required during the operation of the Project.

3.11 Conclusion

3.11.1 Construction Phase

3.11.1.1 Potential air quality impact from the construction works of the Project would mainly be related to the construction activities of site formation, site clearance, excavation, slope works, construction of administration buildings and satellite control buildings, blasting for tunnel construction, mucking out blasted rock and soil from drill-and-blast operations, material handling and backfilling, foundation works, handling and stockpiling of dusty materials, and wind erosion of exposed area. Construction activities of the concurrent projects within 500 m assessment area would also pose cumulative construction air quality impact. With the implementation of mitigation measures specified in the *Air Pollution Control (Construction Dust) Regulation* together with the recommended air quality control measures including frequent watering on active works areas, exposed areas and unpaved haul roads and other site management measures such as good site practices, and EM&A programme, no adverse air quality impact on ASRs would be anticipated during the construction phase.

3.11.2 Operation Phase

3.11.2.1 Cumulative air quality impacts were assessed by considering the Project-induced emissions, all relevant emission sources within the 500 m assessment area, such as vehicular emission from open roads, tunnel portals and concerned facilities with vehicular emissions (public transport interchanges, bus and minibus termini, and heavy goods vehicle and coach parking sites), industrial emissions, and background concentrations. The results concluded that the predicted cumulative air pollutant concentrations on the identified ASRs comply with the respective AQOs during the operation phase. As such, adverse air quality impact due to the operation of the Project is not anticipated.